

The Soul of Transhumanism:
Rationalism and Dissent in the Early History of Transhumanism

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Introduction: A Story About Rationalism, Rabbits, and Cosmocide.

The creature, built like a rabbit but scaled to the size of an elephant, gazed at the clear night sky. A shooting star, puzzlingly slow, streaked through the darkness. The celestial object briefly entranced the gentle giant, before she hopped toward a nearby stream. The long drought that had plagued the plains during the harshest sun-cycle finally relented, and she drank heartily from the cool spring water. Again, she looked up. Because the planet's dim red sun provided so little light, many of the animals had evolved massive eyes for their bodies, in her case, the size of dinner plates. Within them, countless color-sensitive cones gave her pupils the radiant, speckled, and swirling appearance of a far-off nebula. This biological quirk, combined with the evolutionary necessity of eye contact, had fostered a planetwide fascination with the cosmos.

As the light drew closer and stronger, the many eyes of the planet watched the sky, enchanted by the celestial visitor, save for a few sly prey animals that used their predators' distraction to slip away. The rabbitoid, possessing both powerful sight and a large brain, was the first to recognize the object for what it was: a small, cylindrical vessel hurtling through the void. The red sun glinted off the craft's metallic hull, masquerading as a second star. Within hours, the object breached the atmosphere and settled onto the flat ground with a hard, mechanical *thump*. The conversion began instantly. Automated units excavated and sorted resources, replicating themselves; one machine became a legion within a single day. Content with their numbers, the machines began processing the planet like an assembly line, catapulting their plunder beyond the planet's gravity well. The rabbitoid fled for many days, before the ground beneath her caved in.

After the swift harvesting of Planet 4653-B, nothing remained of the endless plains. The machines directed their bounty back to Jupiter to sustain the trillions of digital humans stored

there. The AI governing the solar system recognized that a galaxy-wide campaign was necessary to neutralize any potential threats to humanity, and it pursued this mission with perfect loyalty. Within the endless sprawl of wires, the digital minds remained blissfully unaware of the ongoing cosmocide, locked within an endless series of individually tailored pleasure sequences. In a moral sense, this was the only rational path; the weightless, infinite joy reaped by trillions of digital minds far outweighed the brief suffering of biological byproducts beyond their system.

Understanding Transhumanism

A myriad of transhumanist thinkers inspired this story.¹ But what *is* transhumanism? Transhumanism is rooted in a prefix and suffix : trans, beyond, and human, the biological form of *Homo sapien*. At its core, transhumanism is the belief that humanity should transform itself into something beyond its present limitations. The word "transhuman" itself predates the modern term by centuries, appearing first in Dante Alighieri's *Paradiso* to describe the soul's passage beyond mortal experience, before being repurposed in the twentieth century to describe a secular, scientific project. Central to modern transhumanism, and to the transhumanist tradition examined in this thesis, is rationalism. Rationalism, defined broadly, is the conviction that reason and logical inquiry are the supreme tools for understanding and improving the world, superior to faith, emotion, or tradition. Rationalism is a core part of my analysis within this thesis.

Contemporary transhumanists operate within a small, specific network of organizations where physical augmentation is no longer the sole focus. Instead, the movement now champions

¹ Robert J. Bradbury, "Extropians Forum: Re: Meme Trace: Origin of 'Jupiter-Sized Brains'," Circa 1999, Extropian Forum Archive, <https://diyhlpl.us/~bryan/irc/extropians/www.lucifer.com/exi-lists/extropians.3Q99/2090.html> Robin Hanson, "The Great Filter—Are We Almost Past It?," 1998, <https://mason.gmu.edu/~rhanson/greatfilter.html>. Robert J. Bradbury, "Matrioshka Brains," unpublished manuscript, July 21, 1999, <https://gwer.net/doc/ai/scaling/hardware/1999-bradbury-matrioshkabrain.pdf>. Hans P. Moravec, *Mind Children: The Future of Robot and Human Intelligence* (Harvard University Press, 1988). Nick Bostrom, "Where Are They? Why I Hope the Search for Extraterrestrial Life Finds Nothing," *Technology Review*, May/June 2008, 72–77, <https://nickbostrom.com/papers/where-are-they/>

libertarianism, progression as a mantra, cryonics—the freezing of the human body—strict rationalism, and an obsession with space colonization. These tenets form the core of the contemporary transhumanist movement. Associations with the alt-right and race science have also emerged within recent years. Understanding how this combination of beliefs emerged requires an examination of the historical discourse of transhumanist thought. What did transhumanism look like before it reached this peculiar state, and can pre-1990s transhumanist history recontextualize transhumanism?

This thesis identifies three critical times for transhumanist history: 1901–1910, 1924 to the late 1930s, and the postwar period to 1977. In each of these times, there is a flare up of proto-transhumanist discourse; each flare up was uniquely shaped by the historical context of its time. I examine a number of crucial proto-transhumanist thinkers: H.G. Wells, JBS Haldane, JD Bernal, Bertrand Russell, Olaf Stapledon, Aldous Huxley, Julian Huxley, Arthur C. Clarke, and F.M. Esfandiary. This thesis argues that proto-transhumanist thought was a constantly changing discourse that did not evolve linearly and never possessed any innate traits beyond the changing of the human body into something else, contrary to ideas expressed by prominent rationalist individuals and organizations such as Nick Bostrom, Humanity+, or Less Wrong. Understanding the significance of this argument requires situating these thinkers within the broader historiography of transhumanism.

Period 1 1901–1910: The Edwardian Zenith and the "Long Depression"

The first transhumanist window emerged from a Britain defined by a profound paradox: an outward display of imperial power masked an inward terror of economic and biological decay. Despite the era's perceived prosperity, the nation grappled with the "Long Depression," a period of stagnation where British GDP fell behind the industrial might of Germany and the

United States.² This economic anxiety fueled a "national efficiency" movement, as elites feared Britain's population—the most urbanized in the world—was suffering from "urban degeneration."³ H.G. Wells, a socialite, sociologist, and acclaimed author, famously categorized this underclass as "The People of the Abyss."⁴ This eugenics view of thinking was the norm of Wells' time, and would extend into the interwar period, and even bleed into the post-World War II period through Julian Huxley. For thinkers like Wells, science-based social planning was not a sci-fi fantasy but a social necessity. These early transhumanists imagined a new and improved type of citizen, thriving in a high-tech industrial environment, and utilized socialist frameworks and the burgeoning Labor Party to propose a socialist technocracy.⁵ This early landscape set the stage for transhumanism to evolve from mere eugenics to the mechanical wombs and brain-machine apparatus that was examined in the second period.

Period 2 1924–1939: The Interwar Crisis

The trauma of the "Lost Generation" and the instability of European democracies forged the second window of transhumanist thought.⁶ Following the carnage of World War I, Britain faced a permanent loss of world dominance and an interwar recession that necessitated sweeping change.⁷ As the Great Depression took hold and socialist movements clashed with rising

² Wuyangga, "The Political and Economic Impact of the Long Depression (1873–1896) on Late Victorian Britain," *Journal of Applied Economics and Policy Studies* 18, no. 7 (2025): 53–61, <https://doi.org/10.54254/2977-5701/2025.25766>. J. H. C. Taylor, "Bio-Medical Degeneration in Britain, 1850–1914: Theory and Evidence" (DPhil thesis, University of Oxford, 2021), 5–7, 9–13.

³ Bentley B. Gilbert, "Health and Politics: The British Physical Deterioration Report of 1904," *Bulletin of the History of Medicine* 39, no. 2 (1965): 143–53 Great Britain, Inter-Departmental Committee on Physical Deterioration, Report of the Inter-Departmental Committee on Physical Deterioration (H.M.S.O., 1904), http://archive.org/details/b32180366_0001.

⁴ H.G. Wells, *Anticipations* (Chapman & Hall, 1901).

⁵ Paul Adelman, *The Rise of the Labour Party, 1880–1945*, 3rd ed. (Routledge, 1996). 11–50

⁶ Gero Erdmann and Marianne Kneuer, *Regression of Democracy?*, *Zeitschrift für Vergleichende Politikwissenschaft* (VS Verlag für Sozialwissenschaften, 2011), 247, 259, <https://doi.org/10.1007/978-3-531-93302-3>.

⁷ E. H. Carter, *A History of Britain, Volume VII: Liberal England, World War and Slump, 1901–1939* (Stacey International, 2011).

fascism—most notably in the Spanish Civil War—the "un-augmented" human began to appear biologically obsolete.⁸ Democracy and capitalism were failing, and early transhumanist thinkers like J.B.S. Haldane and J.D. Bernal wrote that science must seize control of the government to better serve all of humanity.

Period 3: 1966–1977 (The Post-War Pivot)

The final historical window moved away from rigid social planning toward a decentralized, geographically fluid movement. This period was defined by the civil rights movement, second-wave feminism, and a growing skepticism toward traditional scientific authority; for instance, the "hippie" movement.⁹ As the geopolitical center of transhumanism shifted from elite English society to globalized networks, early transhumanist figures like the author and philosophy Professor F.M. 2030, or the famous science fiction novelist Arthur C. Clarke embraced and rejected various parts of the eras zeitgeist. This era redefined transhumanism, shifting the focus from state-planned collectivism to the personal, technological, and spiritual transcendence of the individual.

Discussion of scholarly arguments.

There exists a great deal of scholarly texts from both transhumanists and non-transhumanists examining contemporary transhumanism, extending back to study late 80s and 90s transhumanism. However, in terms of a deeper history, stretching from a “beginning point” of transhumanism to the late twentieth century, the field is sparse. There are two main lines of

⁸ Tom Buchanan, *The Impact of the Spanish Civil War on Britain: War, Loss and Memory*, Sussex Studies in Spanish History (Sussex Academic Press, 2007).

⁹ David Kaiser and W. Patrick McCray, *Groovy Science: Knowledge, Innovation, and American Counterculture* (University of Chicago Press, 2019), <https://doi.org/10.7208/9780226373072-001>; Barbara Molony and Jennifer Nelson, eds., *Women's Activism and "Second Wave" Feminism: Transnational Histories* (Bloomsbury Academic, 2017).

thinking, what I call “linear transhumanism,” dominant among transhumanist historians, and what I call “Huxley-centric” transhumanism, dominant among non-transhumanist historians.

The “linear transhumanism” line of thinking overwhelmingly emerges from transhumanist philosophers and transhumanist organizations.¹⁰ This line of thought situates transhumanism’s first stirrings within prehistory, ancient history, or from the Middle Ages to the Enlightenment era. Scholars most commonly use Enlightenment-age rationalism as a marker for congruent transhumanist thought. Rationalism has changed throughout history, but most prominently began in the 1800s as a philosophy advocating intellectual logic and thought as the primary way to understand the world, rather than a different philosophy like faith.¹¹ However, my thesis, as well as other scholars in the field, argues that similar thoughts throughout the ages is not enough to show an intellectual movements history—those are separate intellectual movements that contributed to different fragments of transhumanism but never came close to uniting those fragments in even one thinker, much less a group of people forming a discourse around these ideas.¹²

The second existing argument is that of transhumanism emerging from or through Julian Huxley.¹³ Whether scholars argue he himself came up with the ideas or synthesized them

¹⁰ Nick Bostrom, "A History of Transhumanist Thought," 2005, <https://nickbostrom.com/papers/a-history-of-transhumanist-thought/>. Elise Bohan, *History of Transhumanism* (PhD diss., Macquarie University, 2022), <https://doi.org/10.25949/19444418.v1>. Natasha Vita-More, "History of Transhumanism," in *The Transhumanism Handbook*, ed. Newton Lee (Springer International Publishing, 2019), https://doi.org/10.1007/978-3-030-16920-6_2. "Transhumanism Timeline: Significant Contributions Shaping the Evolution of Transhumanist Philosophy," Transhumanist Party, accessed December 9th, 2026, <https://transhumanist-party.org/transhumanism-timeline/>. "Prehistory of Transhumanism," H+Pedia, accessed November 14th, 2025, https://hpluspedia.org/wiki/Prehistory_of_Transhumanism.

¹¹ Joshua Bennett, "A History of 'Rationalism' in Victorian Britain," *Modern Intellectual History* 15, no. 1 (2018): 63–91, <https://doi.org/10.1017/S1479244315000438>.

¹² Ingrid Dunér, *Julian Huxley, Evolutionism and the History of Transhumanism*, Palgrave Studies in the Future of Humanity and Its Successors (Palgrave Macmillan, 2025), <https://doi.org/10.1007/978-3-031-81720-5>.

¹³ Dunér, *Julian Huxley*, 56–67; Ingrid Dunér, "Huxley's Latest Word," in *Julian Huxley, Evolutionism and the History of Transhumanism* (Palgrave Macmillan, 2025), 225–55; Megan Leverage, "Worlds Apart? Julian Huxley,

through the readings of others, a 1951–1957 emergence of transhumanism through the writings of Julian Huxley is a common argument, as he was one of the first thinkers to use the term “transhuman.” Dante Alighieri is broadly credited with creating the term, but the large time gap between Huxley and Alighieri’s use of the term, as well as Alighieri’s use of the term primarily for religious purposes makes an Alighieri-as-origin argument nigh impossible to make. Julian Huxley as origin was historically an unpopular argument and continues to be unpopular amongst transhumanists.¹⁴ This is because the common sentiment was that Huxley’s transhumanism was too different from the transhumanism defined by Max More, the first figure to explicitly organize transhumanists under a transhumanist banner in the late eighties and nineties. More wrote in a mostly empty space in the late twentieth century and his community of transhumanists, formally called Extropians, became the dominant transhumanism. However, since 2015 a wave of papers and a book have argued for Julian Huxley’s inclusion as the origin for transhumanism on the lines that his idea of transhumanism matches up close in concept to Max More’s transhumanism, simply differing on arbitrary details.¹⁵

The historical study of transhumanism is miniscule. Much of the historical thought has been done primarily for philosophical examinations, with a quick dive into history invoked to support a primary argument. Non-historians have created most of these histories; furthermore,

Transhumanism, and Eugenics," *Contemporanea* 21, no. 3 (2018): 431–58; Alison Bashford, "Julian Huxley's Transhumanism," in *Crafting Humans: From Genesis to Eugenics and Beyond*, ed. Marius Turda (V&R Unipress, 2013), 153–68; Paul Weindling, "Julian Huxley and the Continuity of Eugenics in Twentieth-Century Britain," *Journal of Modern European History* 10, no. 4 (2012): 480–99, https://doi.org/10.17104/1611-8944_2012_4_480; Ingrid Dunér, "Evolutionary Apocalypse," in *Julian Huxley, Evolutionism and the History of Transhumanism* (Palgrave Macmillan, 2025), 165–97.

¹⁴ Bostrom, "History of Transhumanist Thought"; Bohan, "History of Transhumanism."

¹⁵ Dunér, Julian Huxley; Dunér, "Huxley's Latest Word"; Leverage, "Worlds Apart?"; Joseph Wolyniak and Peter Harrison, "History of 'Transhumanism,'" *Notes and Queries* 62, no. 1 (2015): 138–39, <https://doi.org/10.1093/notesj/gjv080>; John Kennedy Phillip, "A Philosophical History of Transhumanism," *Philosophy Now*, February 2024, https://philosophynow.org/issues/160/A_Philosophical_History_of_Transhumanism.

only two works —Bohan's *A History of Transhumanism* and Dunér's *Evolutionism and the History of Transhumanism*—provide a broad historical view of transhumanism, rather than examining a single isolated figure or decade. Bohan's work is extremely chronologically wide, beginning prior to the enlightenment, while Dunér begins with Huxley.¹⁶

My Contributions

This thesis takes a different approach to transhumanist history. This paper studies transhumanism as an intellectual history within elite English society in the 1900s that responds to historical context as a living discourse. This situates transhumanism within the widespread acceptance of evolution and rapid advancement of biological understanding. This is not the first time that transhumanist history has been situated within this time, but it is the first to critically examine how these thinkers interacted and clashed, as social products of their time.¹⁷ The thinkers within chapter one and chapter two of this paper are engaged in conversation with each other and their works; the majority of them had dinners and parties with almost all of the other writers examined in these chapters. The earliest thinkers were all socialist or socialist adjacent, and engaged in socialist politics, often joining the same political organizations. My thesis closely examines this lively intellectual-elite world in English society in relation to transhumanism over decades, demonstrating a back-and forth argument on the ideology of transhumanism. This is an analysis which is currently absent or underexamined in transhumanist histories.

¹⁶ Bostrom, "History of Transhumanist Thought"; Bohan, "History of Transhumanism"; Dunér, Julian Huxley.

¹⁷ All three of these works examine some of these early 1900s thinkers, but they are generally quickly gone over, understood as isolated figures coming up with fragments of transhumanism. Bohan, "History of Transhumanism"; Bostrom, "History of Transhumanist Thought"; James J. Hughes, "The Politics of Transhumanism and the Techno-Millennial Imagination, 1626–2030," *Zygon* 47, no. 4 (2012): 757–76, <https://doi.org/10.1111/j.1467-9744.2012.01289.x>.

Rationalism has been used as a prerequisite for transhumanist thought; it was core to the Extropians and is crucial to modern day transhumanists.¹⁸ In my research, I have found no works arguing against the acceptance of all historical transhumanists as rationalists. My thesis makes the original argument that these thinkers exhibited very different types of rationalism and were not ubiquitously rationalists but varied in their primary beliefs and philosophical frameworks. This disentangles transhumanism from being fundamentally rationalist, inherently creating a more inclusive ideological capacity for transhumanism. Running across all three chapters is an original argument about gender: that gendered rationalism was not incidental to proto-transhumanist thought but structural, embedded from Wells forward and never fully examined in existing scholarship.

A number of figures are also baked into this thesis that are absent or limited in other conceptions of transhumanist history. This thesis incorporates Julian Huxley into the fold of transhumanism but rejects him as a beginning. Before he used the term, his peers and mentors were imagining brains within mechanical casings, bioengineered babies grown from machine-wombs through ectogenesis, and the literal transcendence of humanity in the biological, sociological, spiritual, and intellectual senses. At the same time, his popularization of these ideas as well as the religious element he contributed is critical and worth including in a transhumanist history. This thesis also offers the first sustained treatment of Arthur C. Clarke as a

¹⁸ Max More, "Philosophy of Transhumanism," Humanity+, accessed March 1, 2026, <https://www.humanityplus.org/philosophy-of-transhumanism>; Max More, "Extropy #6," Extropy, 1990, <https://diyhl.us/~bryan/irc/extropians/extropy-magazine/ext6.pdf>; Max More, "Transhumanism: Critical Rationalism, Not Omniscient Reason or Scientism," Extropic Thoughts (Substack), February 4, 2024, <https://maxmore.substack.com/p/transhumanism-critical-rationalism>; Timnit Gebu and Émile P. Torres, "The TESCREAL Bundle: Eugenics and the Promise of Utopia through Artificial General Intelligence," First Monday, April 14, 2024, <https://doi.org/10.5210/fm.v29i4.13636>; Ben Goertzel, "The Extropian Creed," 2000, accessed February 28th 2026, <https://www.goertzel.org/benzine/extropians.htm>; Eliezer Yudkowsky, Rationality: A-Z, LessWrong, n.d., accessed February 17th 2026, <https://www.lesswrong.com/rationality>. The Yudkowsky blog pertains to living a more rational existence.

transhumanist thinker within the proto-transhumanist intellectual lineage, examining how 2001: A Space Odyssey crystallized the movement's religious turn and carried forward the gendered hierarchies of its predecessors into mass culture — making those hierarchies not historical relics but living assumptions.

In the same chapter, this thesis provides an original examination of F.M. 2030 as well. F.M. 2030 is not commonly examined in historical transhumanism, and scholars perceive F.M. 2030 as a visionary of Extropianism and modern transhumanism. The majority of these writings justify his importance and contributions to the mainline canon of transhumanism.¹⁹ This thesis argues that F.M. 2030 was *outside* of the mainline canon, with a distinctly different set of core beliefs and approaches to his transhumanism. Furthermore, this thesis examines the majority of his works as a cohesive ideological framework, which is nonexistent in current transhumanist research. Finally, throughout this thesis, I argue and demonstrate that transhumanist historiography has not been a neutral act of recovery but a selective one, consistently privileging cosmic transcendence over earthly humanitarianism, and rationalist transhumanists over transhumanist critiques of rationalism. This is visible in my analysis of Bertrand Russell, Olaf Stapledon, and F.M. 2030. Understanding which voices were included and which were erased is not merely an academic exercise; it shapes what transhumanists believe its future can contain.

Chapter Structure

This thesis is divided into three chapters, and examines a myriad of authors and their writings. I chose authors by examining transhumanist literature to identify the most utilized core figures with documented connections to one another. The thinkers in these chapters form an intellectual web where thoughts are argued back and forth in their respective times, as well as

¹⁹ Vita-More, “History of Transhumanism.”

passed forward through the decades. I chose most influential and prominent works of these authors *on the topic of transhumanism*.

I begin the first chapter in 1901 with H.G. Wells as well as *A Modern Utopia*. H.G. Wells sets much of the intellectual and ideological background for early transhumanism and dominates the social networks and organizations that early transhumanists inhabit. The chapter moves on to 1924-1929 with JBS Haldane and J.D. Bernal, two central figures of transhumanism, imagining core ideas of transhumanism as well as varying in their relationship to rationalism and science worship. I use Bertrand Russell to demonstrate the dynamism of the early movement. The chapter ends in 1929, leading directly into chapter 2. Historical events like the end of the Victorian era, the early 1900s British economic depression and World War I are examined as influencing these author's writings.

Chapter 2 begins with Olaf Stapledon's *Last and First Men* (1930). I situate Stapledon as a voice of dissent among early transhumanists and explore him as a limited rationalist, as spiritualism and emotional connection trumped rationalism within Stapledon's seminal work. Aldous Huxley's *Brave New World* is examined as one of the most famous texts evolving from the transhumanist movement, and is examined as a semi-rationalist text, with a hierarchical and gendered rationalism similar to that of H.G. Wells. Finally, the chapter concludes with Julian Huxley's influence from earlier transhumanists and synthesis of transhumanism, as well as an analysis of his ideas within the infamous *UNESCO* (1946). Residual trauma from World War I, the global Great Depression, and changing ideas of eugenics are discussed as historical context within this chapter.

Chapter 3 is distinct from chapters one and two. Whereas chapters one and two stay centered within elite English society, chapter 3 moves geographically. F.M. 2030, a little-known

U.N. diplomat turned philosophy professor and minor celebrity, is examined as contrasting many of the implicit beliefs popular in early transhumanism. His in-chapter foil is Arthur C. Clarke, who is examined through his famous work, *2001: A Space Odyssey*. *2001* is understood as following a combination of Wells and Stapledonian ideology, creating an artistic work that spreads a more modern, somewhat Extropian understanding of transhumanist thought. *2001* is the climax of the thesis, where earlier transhumanist thought is reborn into contemporary transhumanism, complete with many of the cultural beliefs attached to modern transhumanism. Arthur C. Clarke is directly linked to a number of proto transhumanists through his autobiography. The civil rights movement, second wave feminism, the hippie movement, and the changing role of science are discussed as historical context in this chapter.

Methods and sources

Each of these writers are examined primarily through their books or, in Julian Huxley's case, manifestos. This approach has difficulties—most of these works are fictional, and close textual analysis and interpretation is necessary. However, there is an essential benefit: The famous texts that other proto-transhumanists would read are examined, rather than analyzing only the bluntest texts that may or may not have been read by other proto-transhumanists. Furthermore, a mixture of personal and direct writings—biographies, letters, and essays—are examined for each author, in order to support and inform interpretations of their more famous works. That most of these authors were located in London in the early 1900s has made reading their biographies for accuracy convenient, as the different biographies confirm the same facts.

Priests and Cyborgs: The Making of Transhumanist Rationalism, 1901–1929

Section I: From Victorian Sentiment to Scientific Futures

In 1895, the Victorian era was drawing to a close. Britain, still the center of a vast global empire, stood between an old social order and a rapidly industrializing world. Victorian culture prized emotional restraint, domestic femininity, and the near-sacred stability of social hierarchy—a cultural atmosphere that was nostalgic, moralizing, and deeply suspicious of radical innovation.²⁰ Industrial and scientific transformation was pulling the modern world in a different direction entirely.

It was at precisely this moment that one of the loudest critics of Victorian sentimentality rose to fame: Herbert George Wells, a South London-born writer whose debut science-fiction novel, *The Time Machine* (1895), catapulted him into literary prominence.²¹ Wells would go on to earn the title "the father of science fiction."²² This chapter, however, positions him as something else: the architect of an ideological landscape that two contemporaries—J.B.S. Haldane and J.D. Bernal—would shift and build upon. Wells himself is not a transhumanist in the full sense, but he crystallized many of the ideological structures that early transhumanism

²⁰ Walter E. Houghton, *The Victorian Frame of Mind: 1830–1870* (Yale University Press, 1963), 218–40, 289–344, 341–80.

²¹ Paul A. Cantor, "The Empire of the Future: Imperialism and Modernism in H.G. Wells," paulcantor.io, May 24, 2020, <https://paulcantor.io/paul-cantor-works/the-empire-of-the-future-imperialism-and-modernism-in-hg-wells>. Cantor describes "Wells refuses to accept Victorian complacency and self-satisfaction... Here is the core of Wells's modernism. Indeed he was as thoroughgoing a critic of Victorian pieties as any of the famous modernists, and shared their contempt for Victorian smugness, as becomes evident in his portrayal of the Time Traveler as an explorer."

²² "HG Wells — Father of Science Fiction with Hopes and Fears for How Science Will Shape Our Future," January 19th 2024, ABC Science Show, <https://www.abc.net.au/listen/programs/scienceshow/hg-wells-author-who-imaged-a-future-based-on-scientific-achievements/13731742>.

utilized and divorced from; his enormous influence on British scientific culture leads to each generation of transhumanist thinkers sometimes mentioning, mostly worshipping, his name.

Wells, Haldane, and Bernal defined rationalism in distinctly different ways, each shaped by the historical forces of his moment. Wells established rationalism as the supreme organizing principle of society, coding it as the domain of a scientific elite and inflecting it with the gendered hierarchies of the Victorian world he grew up in. Haldane stripped that rationalism of its remaining religiosity and centered it around biological evolution, imagining ectogenesis and genetic modification as its tools. Bernal pushed further still, envisioning the merger of human and machine as rationalism's logical endpoint. This chapter asks how these early transhumanist thinkers defined rationalism and valorized science in different ways, furthering our understanding of whether these definitions went forward, mutated, or dead-ended on the road to contemporary transhumanist thought.

Section II: Historiography

My approach aligns with Ingrid Dunér in recognizing H.G. Wells, Haldane, and Bernal as important precursors to Julian Huxley, but I differ in focusing on the discourse and change over time of rationalism. Where Dunér situates Wells as a formative influence over Huxley as a scientist and social planner, this chapter focuses on the role of rationalism within this time period and how it evolved from thinker to thinker. Furthermore, I situate Wells not only a precursor but the historical backdrop in a developing argument about rationalism and science worship that grows more explicit and contested with each generation. Section IV of this chapter argues for unique characterization of this background along the lines of a theological-rationalist system, heavily steeped in Nietzschean influence. This distinguishes my interpretation from those of

Nick Bostrom and Elise Bohan as well, both of whom situate Wells as a speculative or narrative precursor, either as a provider of imaginative scenarios within a broad transhumanist tradition, or as an early voice in a sweeping continuous story of human becoming.

I identify J.B.S. Haldane as a critical turning point: both a transformation of Wellsian rationalism and a credible origin point for transhumanism itself. Bertrand Russell and J.D. Bernal then serve as key interlocutors whose debates and divergences actively reshape transhumanism's rationalist foundations. This differs from Dunér's more continuous intellectual genealogy centered on the Wells–Huxley relationship, and from Bostrom and Bohan, whose accounts do not foreground these figures as central nodes of transformation. Rather than tracing a line of inheritance, this thesis traces a discourse, defined by discontinuity, contestation, and structural change.

As discussed in the introduction, transhumanist historiography divides broadly into two camps: the linear model, which traces isolated proto-transhumanist fragments across centuries, and the Huxley-as-origin model, which locates the movement's beginnings in the 1950s.²³ Both models share a critical blind spot: they treat Wells, Haldane, and Bernal as isolated precursors rather than as participants in an evolving discourse, and they accept the rationalism of these thinkers as stable, rather than examining how differently each defined it. This chapter argues that reading these three thinkers as within a shared discourse, rather than a sequence of isolated figures, reveals something neither model can account for: a rationalism that was not monolithic but historically influenced. Wells receives an extended treatment in this chapter not because he is

²³ For the linear model, Bostrom and Bohan are the best examples. For the Huxley as origin model, Dunér, Alison Bashford's "Julian Huxley's Transhumanism" and Megan Leverage's "Worlds Apart? Julian Huxley, Transhumanism, and Eugenics" are the best examples.

the most radical of these thinkers, but because he is the most foundational, an intellectual backdrop against which Haldane, Bernal, and the thinkers of chapter two, Olaf Stapledon, Julian Huxley and Aldous Huxley, all defined themselves, whether by inheritance, mutation, or rejection.

Section III: Framing Early Transhumanist Discourse

The three thinkers at the center of this chapter were not isolated intellectuals writing in a vacuum. They belonged to an overlapping world of political and scientific institutions, attempted to answer the same problems at the core of British economic depression and societal confusion, and shared a conviction that science-based planning was the ideal path to a better society. H.G. Wells (1866–1946), the eldest of the three, was a dominant public intellectual of Edwardian Britain, a prolific novelist, futurist, and committed socialist.²⁴ JBS Haldane (1892–1964), was born into biology; his father was a biologist, and Haldane became famous even as a boy for his chemical experiments on his own body.²⁵ J.D. Bernal (1901–1971), the youngest and most politically radical, was a highly esteemed crystallographer and committed Marxist, who famously proposed a brain-machine blend.²⁶ They differed and overlapped in their approaches to rationalism, science, and the burgeoning idea of transhumanism. What distinguished these thinkers from mere science fiction writers was that their work was futurology: serious attempts by working scientists and public intellectuals to envision and direct humanity's rational

²⁴ David Clayton) Smith, *H.G. Wells: Desperately Mortal: A Biography* (New Haven : Yale University Press, 1986), n1-n15, 13, 27, 39-41, 58, 63, 82, 91, .

²⁵ Ronald Clark, J.B.S.: *The Life and Work of J.B.S. Haldane* (Oxford University Press, 1984), 27, 59–62.

²⁶ Andrew Brown, J.D. Bernal: *The Sage of Science* (Oxford University Press, 2005), 46–65, 308, 381–421.

transformation, written with the conviction that the future they imagined was not just a fictional fantasy but a blueprint for a better future.

What scientific rationalism meant, and who it was for, shifted considerably across the three thinkers works. Wells conceived of rationalism as inseparable from a particular kind of person: educated, scientifically trained, and implicitly male, inflected throughout with the gendered, elite hierarchies of the Victorian world he grew up in. Wells suffused rationalism with a quasi-religious devotion, framing scientific pursuit as a sacred path toward God. In short, Rationalism carried with it moral implications of a person's character. Haldane and Bernal shared a commitment to scientific rationalism but shed its overtly hierarchical dimensions. They replaced elite social judgment with biological and technological ambition and grounded their visions in a socialist world government that would democratize science rather than concentrate it among elites. They both differed in their extent of science worship; Haldane elevated the scientist to a godlike figure taking the place of theology. Bernal shed theology and God-like status entirely instead believing in an innate and secular humanistic duty to better mankind. The discourse from Wells to Haldane to Bernal is therefore not simply one of escalating technological ambition, but of a rationalism and science veneration gradually untethered from the class, gender, and spiritual hierarchies that had originally defined it.

SECTION IV: Understanding H.G. Wells

Few British intellectuals of the Edwardian era commanded the cultural authority of H.G. Wells. His 1901 collection of essays, *Anticipations*, was a bestseller that catapulted him into the conversations of middle and upper class British society, and earned him invitation to an elite

British organization, the Fabian society.²⁷ *Anticipations* imagined and planned a Utopian future in a uniquely meticulous fashion; answering questions of how to create and maintain Utopia, the role of science within Utopia, and how to deal with the non-elite many saw as unfit for Utopia. In response to *Anticipations*, The New York Times hailed him as "the foremost sociological novelist of the day," and the London News praised his "brilliant manner."²⁸



Figure 1. H.G. Wells. Photograph. Atomic Heritage Foundation.
<https://ahf.nuclearmuseum.org/wp-content/uploads/2015/10/HG%20Wells-1024x1024.jpg>.

²⁷ Margaret Cole, "H.G. Wells and the Fabian Society," in *Edwardian Radicalism, 1900–1914: Some Aspects of British Radicalism*, ed. A. J. A. Morris (Routledge & Kegan Paul, 1974); Smith, H.G. Wells, 92.

²⁸ W. L. Courtney, "The Novels of H.G. Wells," *New York Times*, 1912; "The Ingenious Philosophic Romancer's Latest Imaginative Work," *New York Times*, 1905, BR342. I have included "The ingenious.... Work" to demonstrate the extent of respect for Wells; I also think it is funny.

A Modern Utopia (1905) differed in format to *Anticipations*; it was a novel, where Wells and an acquaintance were transported to a planet remarkably similar to earth, even sporting doppelgangers, but under a utopian world government.²⁹ Despite this difference, *A Modern Utopia* was intended as a sequel to *Anticipations* and continued in its predecessor's wake, sparking the formation of numerous "Samurai societies" created by high class readers who were inspired by Wells' vision of a rational governing elite.³⁰ Wells was received as a novelist, socialist, sociologist, philosopher, and scientist rolled into one commanding public voice. Wells, like Haldane and Bernal, did not write science fiction or vague philosophy, but rather scientific futurism, conceived in response to historical contexts, aimed at changing the world.

A societal crucible forged Wells' futurism; a profound paradox defined the Britain of Wells' era. Britain projected imperial power throughout the world while masking an inward terror of economic and biological decay. The nation had endured the Long Depression—a twenty-six-year period of stagnation during which British production fell behind the industrial output of Germany and the United States. Elites feared that Britain's population, at the time the most urbanized in the world, was suffering from genetic urban degeneration.³¹ Eugenic thinking was not a fringe position in this climate but a mainstream response, one that would persist through the interwar period and only diminish after World War II.³² Wells's response to the crisis English society faced was characteristically ambitious: in 1903 The Fabian Society, an elite gradualist organization who influenced the formation of Britain's modern Labour Party and

²⁹ H. G. Wells, *A Modern Utopia* (Chapman & Hall, 1905), 1–34.

³⁰ Smith, *H.G. Wells*, 101.

³¹ Taylor, "Bio-Medical Degeneration," 150–202, 237, 15–70; Wuyangga, "Political and Economic Impact of the Long Depression," 2–9.

³² Paul Weindling, "Julian Huxley and the Continuity of Eugenics in Twentieth-Century Britain," *Journal of Modern European History* 10, no. 4 (2012): 480–99, https://doi.org/10.17104/1611-8944_2012_4_480.

created the London School of Economics, invited H.G. Wells to join them because of *Anticipations*.³³ Wells found its reformist pace timid and ineffectual. He attempted to seize complete control of the organization and push it toward more radical ends; when he failed, he left in 1908.³⁴

It was this radical, socialist fusion with scientific rationalism that animated both *Anticipations* and *A Modern Utopia*. Across both works, Wells imagined society governed by a new kind of human being—his "Engineers" in *Anticipations*, his "Samurai" in *A Modern Utopia*—an educated, scientifically trained elite whose rational superiority justified not only their leadership but the elimination or management of those deemed unfit. This was not abstract futurism. Wells explicitly imagined a eugenics program administered by his rational elite, writing that those with transmissible diseases or incurable habits of mind "exist only on sufferance" and that his future men "will not hesitate to kill when that sufferance is abused."³⁵ Rationalism, in the Wellsian framework, was not merely an intellectual orientation but a moral one: to be rational was to be good, and to be good was to be rational. Those who could not meet this standard, such as the mentally ill, emotional, superstitious, and feminine, were not simply different but dangerous, a drag on the forward progress of civilization. Wells transmuted rationalism from an ideology into something approaching a religion, framing scientific pursuit as a sacred path toward a higher social order.

³³ Josephine Fishel Milburn, "The Fabian Society and the British Labour Party," *Western Political Quarterly* 11, no. 2 (1958): 319–39, <https://doi.org/10.2307/444409>; "History," London School of Economics, accessed March 19, 2026, <https://www.lse.ac.uk/about-lse/our-history>.

³⁴ Peter Gahan, "1905–9: Noises Off — H. G. Wells Among the Fabians. Votes for Women!," in *Bernard Shaw and Beatrice Webb on Poverty and Equality in the Modern World, 1905–1914* (Palgrave Macmillan, 2017), https://doi.org/10.1007/978-3-319-48442-6_3, 17, 29.

³⁵ Wells, *Anticipations*, 299–300

At the core of Wells's rationalism was something other than political philosophy: theology. Wells did not abandon the spiritual impulse of the Victorian world he grew up in but instead redirected it. Where Victorian faith asked its adherents to follow divine authority unquestioningly, Wells proposed a new kind of sacred commitment: the pursuit of rational, scientific truth as a path toward understanding God. The scientist replaced the priest; the laboratory replaced the church. This transmutation is nowhere clearer than in *Anticipations*, where Wells described the shift from traditional faith to rational deduction through the metaphor of a lantern as faith, illuminating God:

The ray of the lantern indicated and directed and one followed it...But, indeed, with the new light there has come the time for new methods; the time of lanterns, the time of deductions from arbitrary first principles is over. The act of faith is no longer to follow your lantern, but to put it down.³⁶

He transmuted the act of faith into an act of rational inquiry. Whether Wells was writing in a way his religious audience would find tasteful or truly believed this was a new form of faith is unknowable and matters little for this thesis. Ultimately, his most impactful work preserved the devotional intensity of religion while replacing its mechanism with rationalist science. To be rational was not merely to be correct; it was to be righteous and holy. This sacralization of rationalism had significant consequences for how Wells imagined the social order. If rationalism was a form of sacred duty, then those incapable of it were not simply inferior but morally deficient and therefore unworthy of the future Wells was building. The engineer and the Samurai were not just administrators; they were an ordained, secular priesthood, and their authority derived not from birth or wealth but from their capacity for rational thought. This is the sacred foundation on which Wells's hierarchies rest.

³⁶ Wells, *Anticipations*, 294.

The sacralization of rationalism gave Wells's hierarchy its particular edge, and that hierarchy did not exist in a vacuum. It emerged directly from the crisis of urban degeneration that defined Edwardian Britain. Wells had a name for the flailing masses crowding Britain's industrial cities: "The People of the Abyss," a term that captured both horror at poverty and his cold conviction that such people represented a problem to be managed.³⁷ Wells suggested that efforts should be made to help such people, but he ultimately decided that those who could not conform to society and continued to harm it should be executed. The philosophical current running beneath this willingness to kill in the name of the elite is almost certainly Nietzschean. Following the first English translation of Nietzsche in 1896, his ideas spread rapidly through British intellectual circles, as F.S. Thatcher has demonstrated.³⁸ Wells was very likely amongst those directly influenced by Nietzsche or at least shaped by the broader Nietzschean atmosphere saturating elite British culture in these years.³⁹ Wells's *Engineers and Samurai* bear a striking resemblance to Nietzsche's *Übermensch*; both were cast as superior individuals, who would transcend herd morality and impose his will on the future.⁴⁰

Like Nietzsche, Wells had little patience for conventional morality as a constraint on the exceptional man; his rationalists operated beyond ordinary ethical categories, their authority derived not from democratic consent but from innate superiority. In *A Modern Utopia*, he philosophized that "the true objection to slavery is not that it is unjust to the inferior but that it

³⁷ Wells, *Anticipations*, 116, 221, 280, 294, 316.

³⁸ David S. Thatcher, *Nietzsche in England 1890–1914: The Growth of a Reputation* (University of Toronto Press, 1970), 6–31.

³⁹ Thatcher, *Nietzsche in England*, 82, 83, 170, 209, 279. Notably, Thatcher never finds a 'perfect' source saying Wells read Nietzsche. With that being said, he identifies a number of closely related themes, the widespread nature of Nietzsche in Elite English society at this time, as well as a primary source from the late 1800s comparing Wells to Nietzsche in writing.

⁴⁰ Wells, *Anticipations*, 104–30.

corrupts the superior.”⁴¹ Those deemed detrimental for society existed, in Wells's formulation, "only on sufferance," and his idealized, rational men would have "little pity and less benevolence" for those who failed to meet the rational standard.⁴² Eugenics, in this framework, was the logical expression of rationalism, the mechanism by which the People of the Abyss would be gradually eliminated from the gene pool of a rational civilization.

Wells's rationalism did not exist in a gender-neutral state. The irrationalism he feared and despised was, in the cultural logic of his time, fundamentally feminine. Scholars have observed that emotional restraint was understood as a male virtue and emotional expression as a female failing across the period.⁴³ Darwin himself crystallized this assumption when he wrote that unlike "children, women, savages, and the insane, Englishmen rarely cry."⁴⁴ Wells did not invent gendered rationalism, but he futurized it., understanding the role of gender in one of the core parts of their ideology is important.⁴⁵

Scholars of Victorian gender, including Philippa Levine and Joanne Begiato, have documented how emotional restraint was coded as masculine and emotional expression as feminine across the period, but neither examines how Wells carried these assumptions forward into his futurist writing. Wells scholarship, represented most thoroughly by David Smith's biography, addresses his socialism and scientific futurism at length without examining how gender structured his conception of rationalism. Transhumanist historians — Bostrom, Bohan,

⁴¹ Wells, *A Modern Utopia*, 324

⁴² Wells, *Anticipations*. 299-300

⁴³ Joanne Begiato, "Punishing the Unregulated Manly Body and Emotions in Early Victorian England," in *The Victorian Male Body*, ed. Joanne Ella Parsons and Ruth Heholt (Edinburgh University Press, 2018); Houghton, *The Victorian Frame of Mind*. 218-240, 289-344.

⁴⁴ Thomas Dixon, *Weeping Britannia: Portrait of a Nation in Tears* (Oxford University Press, 2015), 197.

⁴⁵ Hank Pellissier, "Transhumanists: Who Are They? What Do They Want, Believe, and Predict? (Terasem Survey, Part 5)," Institute for Ethics and Emerging Technologies, September 9, 2012, . Abou Farman, "Transhumanism," in *The Open Encyclopedia of Anthropology*, ed. Felix Stein (2022), <http://doi.org/10.29164/22transhumanism>.

and Dunér — similarly leave the gendered dimensions of proto-transhumanist rationalism unexamined, treating these assumptions about gender background rather than as subject of analysis. There is currently no literature examining how gender roles interacted with Wells' conception of rationalism, or how gender shaped the conception of rationalism within transhumanist history more broadly. As modern transhumanist membership continues to skew overwhelmingly male, understanding how gender was embedded in one of the movement's core ideological commitments from its earliest stages is not merely a historical footnote; it is a contribution to understanding why that skew persists. This thesis traces how that gendered rationalism dissolved and mutated across successive thinkers, folding into my broader change over time argument regarding rationalism in transhumanist discourse.

This gendered rationalism is visible in *A Modern Utopia*, where Wells structures the entire narrative around a gendered contrast. His traveling companion, an overly emotional, lovesick biologist, functions as a symbol for everything Wells finds contemptible: an irrational man, tedious, vapid, and incapable of rising above feeling to thought.⁴⁶ The biologist is ultimately rejected by his love interest's doppelganger on the utopian planet, who chooses instead a rational Samurai, the doppelganger of the man she had already chosen over the biologist when they were on Earth.⁴⁷ The message is not subtle; Rationalism is attractive whereas sentimentality is repellent. And while the foil is male, the coding is explicitly gendered; feeling without thought

⁴⁶ Wells, *A Modern Utopia*, 341. The biologist being male is crucial to the plot of the story, as a woman would be in a fundamentally different position of being pursued rather than pursuing, as well as having less autonomy. It could be argued that Wells made the biologist a man to show that this is not gendered, but there is no evidence for this. His exact motive is unknowable, but I find this line of thought highly improbable.

⁴⁷ Wells, *A Modern Utopia*, 339-341

was, in Wells's framework, a feminine failing that any man who aspired to the Samurai class had to overcome.

Women themselves occupy a tightly circumscribed role in Wells's rational society. In a lesser-known 1922 work, *The Secret Places of the Heart*, Wells bluntly outlines his opinions on women. Wells writes through a rational, intellectual characters eyes “We have to make women more responsible.... as guardians and helpers of men... suppress them as tempters... instead of mothering babies they have to mother the race...”⁴⁸ Wells imagined women as literal mothers, schoolteachers, and demonstrators of feminine virtue, their reproductive and domestic labor consciously directed toward producing the next generation of rational engineers. Their sphere was separate, but it was an active sphere, not a passive one, functioning as a service line feeding the rational elite. As Philippa Levine, a Professor on Britain and expert on gender within the Victorian era, shows in *Victorian Feminism*, the ideological separation of men and women into distinct spheres was common in Victorian and Edwardian thought but rare in actual day to day practice.⁴⁹ *Anticipations* and *A Modern Utopia* propose a series of concrete mechanisms that actualize this separation: a more ordered environment, technology reducing the burden of housework, and the strict limitation of frivolous spending, particularly the feminine kind, under a socialist framework that would remove the corrupting influence of consumer capitalism from domestic life.⁵⁰ Wells, regardless of intention, created a futurist solution to intensify aging gender roles.

⁴⁸ H. G. Wells, *The Secret Places of the Heart* (Macmillan, 1922), 92.

⁴⁹ Philippa Levine, *Victorian Feminism, 1850–1900* (Hutchinson Education, 1987), 5-30.

⁵⁰ Wells, *Anticipations*. 103-141

A traditional source of female moral authority in Victorian culture was the idea of innate feminine purity. Levine notes that women's natural virtue made them the moral guardians of the home and society.⁵¹ Wells systematically dismantled this authority, arguing that women's purity had been corrupted by shareholders and capitalist consumer culture, rendering them liabilities rather than anchors. The irony was sharp. Pre-suffragette feminists in Britain had found their most viable political home in socialism and the burgeoning Labour Party, institutions Wells was embedded in.⁵² Wells' works rejected the feminist project those institutions were increasingly attracting. Where feminism in this period meant stepping outside of existing gender norms, Wells deployed the futurist language of technology and rational planning to enshrine those norms.

The assumptions Wells embedded in transhumanism did not vanish as the discourse evolved. They echo, sometimes strikingly, in the transhumanist discourse that emerged nearly a century later. The Nietzschean current running through his *Engineers and Samurai* resurfaced explicitly in the Extropian movement: Max More declared that "transhumanist ideas were directly influenced by Nietzsche," and Extropy magazine, the movement's founding publication, presented its readers with the *Übermensch* as an aspirational ideal.⁵³ Another example is the quasi-religious intensity with which Wells sacralized rationalism finds its echo in what scholars have identified as transhumanism's function as a "secular faith."⁵⁴ Transhumanism substitutes

⁵¹ Levine, *Victorian Feminism*, 13-15.

⁵² Levine, *Victorian Feminism*, 16-18.

⁵³ Jules Evans, "How Did Transhumanism Become the Religion of the Super-Rich?," *Philosophy for Life*, October 28, 2022, <https://www.philosophyforlife.org/blog/how-did-transhumanism-become-the-religion-of-the-super-rich>; Max More, "The Overhuman in the Transhuman," *Journal of Evolution and Technology* 21, no. 1 (2010), <https://jetpress.org/v21/more.pdf>.

⁵⁴ Hava Tirosh-Samuelson, "Transhumanism as a Secularist Faith," *Zygon* 47, no. 4 (2012), <https://doi.org/10.1111/j.1467-9744.2012.01288.x>; Roy Jackson, "Secular Transhumanism as Scientism?," in *Muslim and Supermuslim: The Quest for the Perfect Being and Beyond*, ed. Roy Jackson (Springer International Publishing, 2020), https://doi.org/10.1007/978-3-030-37093-0_3 231-270.

technological immortality and artificial omniscience for the theological promises it officially rejects.

Section V: JBS Haldane and *Daedalus*

The thinker most responsible for transforming Wells's sweeping social vision into concrete biological proposals was John Burdon Sanderson Haldane. Where Wells had imagined a broad future shaped by rational elites, Haldane asked a more precise question: what, exactly, could science do for the human body? Born into the British scientific aristocracy, Haldane's father was John Scott Haldane (1860-1936), a distinguished physiologist who famously used his own son as an experimental subject in studies of gas mixtures and decompression.⁵⁵ J.B.S. Haldane was ever the enthusiast for these experiments, he came to scientific radicalism early and never left it. He volunteered in World War I as a British Army officer, experiencing firsthand the industrialized slaughter that shattered his generation's remaining faith in Victorian sentimentality.⁵⁶ Eric Hobsbawm captured the cultural aftermath precisely: after the carnage, "appeals to emotion rang hollow; what the world trusted now were experts."⁵⁷ Haldane was the embodiment of that shift, a man in whom the war's destruction of old certainties had cleared space for a purely scientific worldview.

In this thesis, Haldane is examined in a new light; he fundamentally changed the rationalism-science-religion paradigm Wells set up, to instead oust religion while creating a hierarchy placing science over rationalism. Haldane also sheds some of the gender infused into

⁵⁵ Clark, *J.B.S.*, 65.

⁵⁶ Clark, *J.B.S.*, 22, 45, 157.

⁵⁷ E. J. Hobsbawm, *The Age of Extremes: A History of the World, 1914–1991* (Vintage Books, 1996), 257.

Wells' rationalism. This change in rationalism is nonexistent in transhumanist histories and the crux of this thesis's argument.

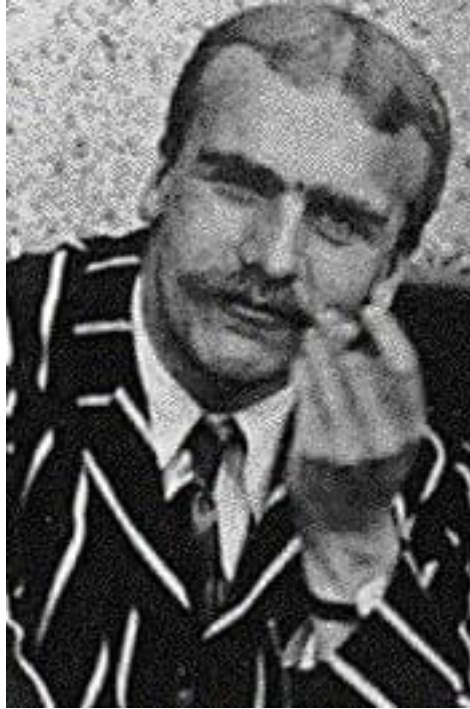


Figure 2. J.B.S. Haldane. Photograph. Sourced from Ranker.com. https://imgix.ranker.com/node_img/67/1339011/original/j-b-s-haldane-writers-photo-1.

Haldane differed from Wells on the fundamental question of who science existed for. Wells had imagined science as the exclusive province of a rational elite, his Engineers and *Samurai* administering the future from above while the People of the Abyss were managed or outright executed. Haldane's vision was structurally different. He believed science belonged to everyone, a conviction he put into institutional practice. Haldane contributed a weekly science column to the *Daily Worker* from 1937 onward and served as chairman of the editorial board of

the publication, Britain's Communist Party newspaper, from 1940 to 1950.⁵⁸ Haldane joined the British Communist Party in 1942, remaining a member until 1950, when he resigned over the Lysenko controversy.⁵⁹ The Lysenko controversy refers to the Soviet endorsement of a pseudoscientific theory of heredity, resulting in the imprisonment and executions of thousands of soviet biologists that did not embrace Lysenkoism. His commitment to socialism was hopeful and humanistic: science as in service of people rather than people in service of science. This distinction is crucial for understanding the nature of proto-transhumanism discourse as chaotic and fluid.

The most immediate way in which Haldane departed from Wells was theological. Where Wells had preserved the spiritual intensity of Victorian culture by recasting scientific pursuit as a path toward God, Haldane rejected the premise entirely. He retained genuine admiration for Jesus as a moral figure and psychologist—going so far as to call Einstein "the greatest Jew since Jesus" in *Daedalus*—but made clear that this admiration extended no further than the human.⁶⁰ He makes this *extremely* clear when he proclaims that "there can be no truce between science and religion."⁶¹ This was not mere secularism but a fundamental reorientation of the scientist's relationship to authority. Where Wells had utilized a model of a devout rational pilgrim moving closer to God through science, Haldane found his in the figure of Daedalus—the craftsman of Greek mythology who built the Minotaur's labyrinth, pioneered experimental genetics, and invented flight, all while remaining entirely indifferent to divine judgment.

⁵⁸ Clark, *J.B.S.*, 122-129.

⁵⁹ Clark, *J.B.S.*, 168-190.

⁶⁰ J. B. S. Haldane, *Daedalus or Science and the Future* (Kegan Paul, Trench, Trubner, 1924), 11.

⁶¹ Haldane, *Daedalus*, 90.

Haldane lingered on the significance of Daedalus' fate: "Despite causing the death of Zeus's son] he was neither smitten by a thunderbolt, chained to a rock, nor pursued by furies... We can hardly imagine him as a member of the throng of shades... He was the first to demonstrate that the scientific worker is not concerned with gods."⁶² This was Haldane's scientist: Secularly sovereign, operating entirely outside divine jurisdiction, subject to no authority beyond the fundamental rules of science itself. Where Wells had transmuted rationalism sacred, Haldane dissolved the sacred entirely and caused rationalism to orbit around science. In this way, Haldane created a science first, rationalism second framing. Haldane's biologist did not answer to God, to morality, or Victorian gender norms. Instead, the scientist answered only to what science could do.

The centerpiece of *Daedalus* was a proposal that marked a genuine threshold in proto transhumanist thought: ectogenesis, the conception and gestation of embryos outside the human body in a mechanical womb. Haldane imagined it initially as a practical solution to declining birth rates, gradually adopted until it became the dominant mode of human reproduction. The eugenic logic was explicit: the most genetically fit individuals would supply the genetic material, and the mechanical womb would create a better human. But the significance of ectogenesis extended well beyond its eugenic mechanics. This could be argued as a beginning point for true transhumanist thought. While it is an endless philosophical argument what the exact limits of transhumanist thought are—some authors have pointed to the use of the most basic tools, such as sticks, as transhumanist—this is the first explicit human modification within this time period.⁶³

⁶² Haldane, *Daedalus*, 48.

⁶³ Bostrom, "History of Transhumanist Thought"; Max More, "The Philosophy of Transhumanism," in *The Transhumanist Reader: Classical and Contemporary Reflections on the Science, Technology, and Philosophy of the Human Future*, ed. Max More and Natasha Vita-More (Wiley-Blackwell, 2013), 3–17.

Haldane did not stop at reproduction. Elsewhere in *Daedalus* he speculated on the possibility of glandular injections to help individuals regulate their emotions and to eliminate menopause for the benefit of women.⁶⁴ *Daedalus* also speculates on the possibility of changing the sex-ratio.⁶⁵

The gendered implications of *Daedalus* are largely accidental. Haldane himself made no explicit judgments about women in the text beyond a passing observation that they were scientifically minded, using the evidence that they had embraced the hygienic demands of germ theory more readily than men. This stands in stark contrast to Wells, for whom the subordination of women was theorized and embedded in the rational society he imagined. Haldane framed the elimination of menopause through glandular injections as a form of biological equity rather than a mechanism of control. Still, his glandular proposals carried implications Haldane did not examine. Within a subculture still saturated with Victorian assumptions about emotion, femininity, and rationalism, the idea of chemically reducing emotional and reproductive biology carried an unmistakable undertow: the body most associated with irrationalism in that cultural framework was the female one. Furthermore, Haldane's advocacy for a sex-ratio implied women's essentiality, but not necessarily their equality. Ultimately the gender-inequity nature of these proposals is in their implications. Regardless of Haldane's intent, *Daedalus* is a far-cry from the hierarchical worldviews within *Anticipations* and *A Modern Utopia*, strong evidence for a rapidly changing discourse. Distant from either of these worldviews was a figure often forgotten in transhumanist histories: Bertrand Russell.

⁶⁴ Haldane, *Daedalus*, 70–77.

⁶⁵ Haldane, *Daedalus*, 68–70.

Section VI: Bertrand Russell and the Human Heart

No figure in the early transhumanist orbit was better positioned to challenge its assumptions than Bertrand Russell. A philosopher, mathematician, and public intellectual of massive reputation, Russell moved through the same institutional and social world as Wells and Haldane. He joined the Fabian Society two years after Wells in 1903 and contributed to many of the same journals as Haldane and Bernal. Furthermore, he maintained a decades-long friendship with Wells, who addressed him affectionately in letters as "Bertie."⁶⁶ Russell's rationalism was tempered by a skepticism that Wells and Haldane lacked. Russell held a conviction, sharpened by his arrest and imprisonment for pacifist activism during World War I, that science divorced from kindness and moral feeling was not progress but catastrophe. Where Wells had treated emotion as a feminine liability and Haldane had proposed chemically reducing it, Russell insisted that the "heart"—by which he meant the sum total of humane impulse—was not an obstacle to a rational society but its only possible foundation.⁶⁷ It was this conviction that brought him into direct, intentional collision with Haldane's *Daedalus* in 1924, producing a significant critique for early transhumanism.

⁶⁶ Bertrand Russell, "H.G. Wells," in *Portraits from Memory and Other Essays* (Simon and Schuster, 1956).

⁶⁷ Bertrand Russell, *Icarus, or The Future of Science* (E. P. Dutton, 1924), 6.

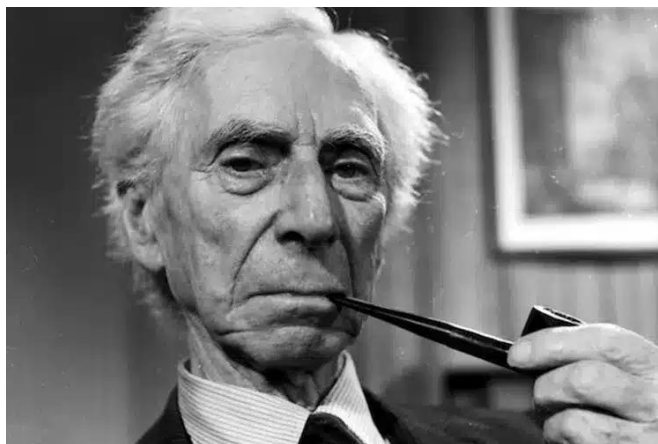


Figure 3. *Bertrand Russell*, photograph by Bassano, 1936.
 Courtesy of National Portrait Gallery, London, via Observer Voice,
<https://observervoice.com/wp-content/uploads/2023/02/Bertrand-Russell-jpg-webp.webp>

When Haldane delivered his early version of *Daedalus* to the Cambridge Heretics Society in 1923, Russell was in the audience. What he heard alarmed him. His short response, *Icarus: or The Future of Science*, published the following year, invoked the myth of the boy who flew too close to the sun on wax wings and fell into the sea. Russell reimagined Haldane's proposals as humanity's wax wings, thrilling in ambition, yet catastrophic in implication. Russell's fear was precise: eugenics and glandular manipulation in the hands of an elite would not liberate humanity but stratify it permanently, engineering subservience into those at the bottom and concentrating genius at the top. His deeper objection ran beneath even this. Russell believed humanity was, at its core, profoundly irrational, not as a failing to be corrected but as a condition to be reckoned with, and that science left this condition unchanged. "Where kindly impulses exist," he wrote, "science helps them to be effective; where they are absent, science only makes men more cleverly diabolic."⁶⁸ The heart, he insisted, was as important as the head. The newspapers were largely unimpressed. *Icarus* was received with a mixture of boredom and disdain, some reviewers expressing open disappointment in Russell for the piece. This reception is itself analytically revealing. By 1924, the cultural authority of the laboratory had so thoroughly

⁶⁸ Russell, *Icarus*, 27–28.

displaced the authority of Victorian sentimentality that even a Nobel laureate's warning registered as regressive. Russell had identified something real: that transhumanism had no internal mechanism for ensuring that its scientific sovereignty would be exercised with kindness.

Transhumanist histories have largely forgotten Bertrand Russell: the forgetting is itself revealing. Bohan's *A History of Transhumanism* dispatches him in a single sentence as posing "counter sentiments," without elaboration. Bostrom's *A History of Transhumanist Thought* is similarly brief, but goes further, effectively erasing any record of his protest. Dunér engages Russell more directly but situates him narrowly as a biographical counterpoint to *Daedalus* rather than as a figure whose critique exposes something structural about the rationalist assumptions transhumanism inherited. This omission is not incidental. Russell's central argument — that humanity's moral and emotional immaturity made it unfit to wield the technologies proto-transhumanism was proposing — represents one of the oldest and most persistent lines of criticism the movement has ever faced; for a modern instance, see Susan Levin's *Posthuman Bliss?*⁶⁹ By writing Russell out of the history, or flattening him into a footnote, transhumanist historians have removed the earliest and most internally credible version of the limited-morality argument, reshaping the past. Transhumanist discourse was not a peaceful river of agreement moving in one direction, but rather a chaotic whirlpool of argument.

Section VII: The Sage of Science: J.D. Bernal

J.D. Bernal's *The World, the Flesh, and the Devil* (1929) arrived quietly. Where *Daedalus* had generated over fifty reviews within a year of publication, *The World* attracted a handful at the time of its release. The reviewer for the Cincinnati Enquirer, one of the few contemporaries

⁶⁹ Susan B. Levin, *Posthuman Bliss? The Failed Promise of Transhumanism* (Oxford University Press, 2021).

to engage with the text at length, grappled openly with the tension between fascination and horror.⁷⁰ Bernal's ideas were creative and intellectually dazzling, but the vision they pointed toward—a humanity that had abandoned its biological body, its home planet, its soul, and its God—produced a vertigo that scientific enthusiasm alone could not overcome. Where Russell strayed too far from the science-forward zeitgeist of the time, Bernal delved deeper than others could follow. It would take decades for the wider world to catch up. A future transhumanist eventually recognized what contemporary reviewers had not: Arthur C. Clarke, whose own work would carry early transhumanism across the threshold into the transhumanism of the late twentieth century, described *The World* as "the most brilliant attempt at scientific prediction ever made."⁷¹ *The World* extended the transhumanist project further than any of its predecessors.

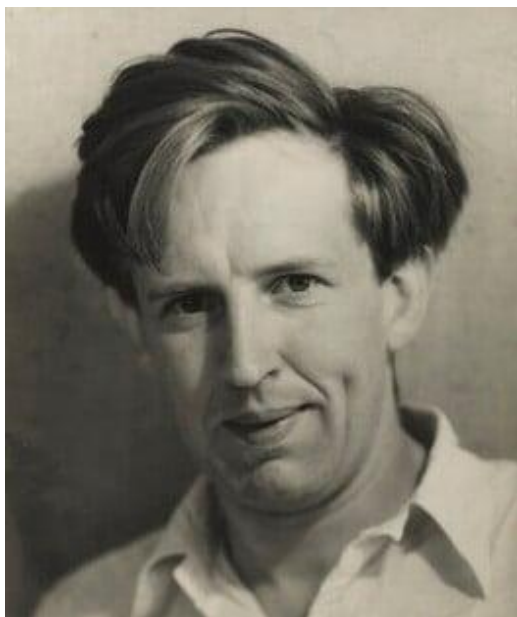


Figure 4. John Desmond Bernal. Photograph. © National Portrait Gallery, London.
<https://collectionimages.npg.org.uk/std/mw08572/John-Desmond-Bernal.jpg>

⁷⁰ "Man of the Future," *Cincinnati Enquirer*, October 8, 1929, Vol 86 Issue 281.

⁷¹ Arthur C. Clarke, *Greetings, Carbon-Based Bipeds!: Collected Essays, 1934–1998* (St. Martin's Griffin, 2000); Brown, J.D. *Bernal*, 70.

Like Haldane before him, Bernal had no patience for theology. Where Haldane had speculated about humanistic hopes for science, Bernal articulated this explicitly in a required, secular sense of humanistic duty. The very title of his work announced his philosophical position. *The World, the Flesh, and the Devil: An Enquiry into the Future of the Three Enemies of the Rational Soul* did not merely invoke rationalism, but enshrined it on the cover, framing the body, spirituality, and human irrationality as obstacles to be overcome rather than conditions to be accepted. The focus of transhumanist discourse drifted away from the science worship of Haldane, back to rationalism. Bernal's rationalism lacked the cold arrogance of Haldane's sovereign scientist or the quasi-religious fervor of Wells's sacred engineer. His scientist was not a godlike figure operating beyond the scope of God, but a perfectly ordinary person with a happenstance talent for science, whose gift conferred responsibility rather than superiority. Science, for Bernal, was *fundamentally* in service of people: a tool, not a throne. This humanism surfaced most clearly in *The Devil*, the section of the work concerned with the future of human society and the obstacles that human irrationality placed in its path.

Bernal acknowledged frankly that humanity was not yet capable of true rationalism, writing that rationalism had "never succeeded" in becoming the dominant partner in human psychology, "not only because its standard was too high to allow any outlet for the primitive forces, but because it was itself too arbitrary, too tainted with distorted primitive wishes ever to be brought into correspondence with reality."⁷² This was a striking, Russell style admission. Bernal's solution was equally striking: a biological and mechanical transformation, channeling humanity into metallic forms better suited to handle emotions while remaining rational. Haldane's ectogenesis and glandular proposals were the direct inspiration for this vision; their

⁷² J. D. Bernal, *The World, the Flesh and the Devil* (Kegan Paul, Trench, Trubner, 1929), 67–68.

logic extended from reproduction and aging into the fundamental architecture of human consciousness itself.

The most radical proposal in *The World, the Flesh, and the Devil*—and arguably in all of transhumanist thought—appeared in *The Flesh*, Bernal's section on the future of the human body. Building directly on Haldane's ectogenesis and glandular proposals, Bernal envisioned something far more total: a melding of the biological body with a mechanical one. He imagined a future human as essentially a brain suspended in a mechanical casing, equipped with artificial sense organs and manipulative appendages that far exceeded the capabilities of biological limbs, connected to other such beings through direct neural communication, living in asteroids converted to space-habitats. This was not science fiction, but futurology offered with the same conviction that had animated Wells's *Engineers* and Haldane's ectogenesis. It was also the moment transhumanism crossed its most significant threshold yet: the human body itself was not merely to be modified or reproduced differently, as Haldane had proposed, but abandoned entirely in favor of a mechanical form better suited to rational existence and space colonization. The idea of the cyborg, the merger of human consciousness with mechanical organs, entered not just transhumanist discourse but world discourse here, in a small barely reviewed book, and it has never left. Bernal acknowledged that most people would find such a transformation horrifying, and that it was entirely understandable that the majority of humanity would initially refuse it, a generosity of spirit toward the unconvinced that was absent in Wells and Haldane's work.

Bernal's political vision, laid out in *The World*, was the most explicitly socialist of the three thinkers. Bernal, envisioned a world government that arrived through incremental

accumulation of scientific knowledge. Bernal theorized as science became more specialized and politicians and the public understood it less, an advisory board of scientists would gradually assume greater power, eventually producing a technocratic socialist worldwide state.⁷³ Crucially, Bernal did not imagine this as tyrannical, because his scientist was not Wells's superior being or Haldane's sovereign figure, but simply a person with a talent for science and no other distinguishing characteristics, not even an above average intelligence. A scientific government that elevated such ordinary people was not anymore a threat to humanity than any other government, and instead an expression of its collective rational capacity.

This vision was not merely theoretical. Bernal and Haldane were core members of what historian of British history Gary Werskey termed the "Visible College," an informal network of British scientific socialists active in the 1920s to the 1940s.⁷⁴ Both Bernal and Haldane were active in the Association of Scientific Workers, both joined the British Communist Party, and both contributed extensively to the *Daily Worker*, where Haldane served as chairman of the editorial board and Bernal as a frequent contributor. Letters held in the Wellcome Collection show them editing each other's work.⁷⁵ The organizational expression of their shared vision was the Social Relations of Science movement, which emerged in the 1930s in response to the Great Depression, rising fascism, and the apparent success of Soviet scientific planning. Bernal's later work, *The Social Function of Science* (1939) served as the movement's manifesto, and historians have noted that Bernal and Haldane "seemed almost to dominate the British scientific world."⁷⁶

⁷³ Bernal, *World, Flesh and Devil*, 87–90.

⁷⁴ Gary Werskey, *The Visible College: The Collective Biography of British Scientific Socialists of the 1930s*, Allen Lane, 1978.

⁷⁵ J. D. Bernal to J. B. S. Haldane, January 1944, Haldane Papers, Wellcome Collection; J. D. Bernal to J. B. S. Haldane, March 1945, HALDANE/4/20/1/27, Wellcome Collection, <https://wellcomecollection.org/works/p2sat5p3/items>.

⁷⁶ Brown, *J.D. Bernal*. 90-95

Section VIII: A Time of Discourse

What emerges from this chapter is not a story of progress but a story of discourse. Wells, Haldane, and Bernal did not constitute a transhumanist movement. They were a handful of individuals, embedded in overlapping social and historical worlds, whose ideas sometimes intentionally and sometimes unintentionally spoke to and built upon one another in ways that are only clear in hindsight. Wells crystallized the scientist as social planner landscape; Haldane and Bernal crossed into genuinely transhumanist territory, the mechanical womb and the cyborg marking the thresholds that Wells's social vision never reached. The de-hierarchicalization of rationalism across these three thinkers was real but unintentional, the residue of different men thinking different thoughts in different times. Transhumanism has never possessed any innate traits beyond the transformation of the human body into something else, and the rationalism that has been treated as its stable foundation was, in practice, a constantly renegotiated variable, differently defined by each thinker and shaped at every turn by the historical moment he inhabited. The thinkers of chapter two, Stapledon, Aldous Huxley, and Julian Huxley, continued this contested discourse.

Stapledon, the Huxleys, and the Contested Soul of Transhumanists (1930-1946)

Section I: Transhumanist Writings and Intellectual Aristocracy

London, 1931. We might imagine a room thick with cigarette smoke and the scent of strong English tea, punctuated by three men bent over a manuscript. H.G. Wells, at sixty-five, gestured emphatically at a passage in Julian Huxley's latest draft. "Too timid, Julian. You must be bolder—the world needs planning, conscious direction. The masses cannot guide themselves." Julian, forty-four and already recognized as one of Britain's foremost biologists, nodded thoughtfully while his younger brother Aldous sat slightly apart, his poor eyesight forcing him to lean close to his own papers. The brothers were innately close, but they also had a familial connection to Wells: Wells had studied under their grandfather, T.H. Huxley, "Darwin's bulldog" who died in 1895 when Julian was eight and Aldous barely one, at the Normal School of Science.⁷⁷ The brothers had come to discuss Aldous's forthcoming novel—a dystopia of planning taken to its logical extreme—but the conversation had turned, as it always did in Wells's presence, to the question of how rational men might reshape civilization. "The Samurai," Wells insisted, invoking his favored term from *A Modern Utopia* "We need a new elite, scientifically trained, who will have the courage to make the hard choices."⁷⁸

We might also imagine two hundred miles north, in a modest Liverpool study overlooking the Mersey, Olaf Stapledon sitting alone at his desk. The philosophy lecturer had

⁷⁷ H. G. Wells, *Experiment in Autobiography* (Macmillan, 1934), 159, 617.

⁷⁸ Juliette Huxley, *Leaves of the Tulip Tree* (John Murray, 1986), 120–61; Julian Huxley, *Memories* (Harper & Row, 1970), 139–49. This as well as the next paragraph are an imagined recounting of how one of the many meetings between these men focused on social planning may have occurred. This is not based on any specific historical document and should be taken as fictional writing. The sources provide evidence of these three men writing in different sets of pairs together, but not a specific instance where all three of them wrote together. Still, given Aldous Huxley's presence in Julian Huxley's life as he directly collaborated with H.G. Wells on *The Science of Life Itself*, it is possible, if not probable, that the three men discussed works together.

spent years wrestling with the same questions that animated the London gathering: humanity's future, evolution's trajectory, and the role of reason in remaking the world. But Stapledon's vision, slowly taking form in the manuscript before him, differed fundamentally from the technocratic certainties of Wells and the Huxleys. In his *Last and First Men*, rationality was not humanity's apex but merely a stage, necessary, but ultimately insufficient. His final species, the Last Men, achieved greatness not through perfecting rationalism or intellectual capacity but by transcending these traits, wedding scientific knowledge to an endless spiritual love. Where Wells imagined elites managing the masses, Stapledon wrote of species-wide consciousness, sparking evolutionary change not through planned breeding but through spiritual choice.

The two visions, one of rational control, one of spiritual transcendence, emerged from the same British intellectual milieu, shaped by the same post-war anxieties, yet arriving at profoundly different conclusions and hopes about what humanity might become. This chapter traces how these thinkers imagined the wonderful and horrific implications of the transhumanist ideas their predecessors had proposed through fiction and institutional planning, arguing that their different perspectives shows that discourse varied not just in what transhumanism was in a scientific sense, or how society would be organized or improved, but the existential implications for humanity of transhumanism.

The thinkers examined in this chapter — Olaf Stapledon, Aldous Huxley, and Julian Huxley — have received uneven and largely compartmentalized treatment in existing scholarship. Stapledon has been examined extensively within science fiction studies, most notably by Robert Crossley in *Olaf Stapledon: Speaking for the Future* (1994), which provides the most thorough biographical and literary treatment of his work, and by Patrick Parrinder, whose work traces Stapledon's relationship to Wells. Neither examines Stapledon as a proto-

transhumanist thinker in any sustained way, and transhumanist histories have largely excised him altogether: Bohan's *A History of Transhumanism* omits him entirely, while Bostrom reduces him to a brief mention of speculative technology. Dunér's *Julian Huxley, Evolutionism and the History of Transhumanism* mentions Stapledon only in passing, in the context of his brief literary resurgence in the 1980s, rather than as a substantive figure in the proto-transhumanist tradition. Aldous Huxley's *Brave New World* has generated an enormous body of literary criticism and is frequently invoked in transhumanist and bioethical debates, but almost always as a dystopian warning sign rather than as a text whose internal hierarchies reveal the ideological assumptions of the tradition that produced it. David Bradshaw's *The Hidden Huxley* remains the most rigorous historical examination of Aldous Huxley's political and social thought, and this thesis draws heavily on it — while departing from Bradshaw's occasionally excusatory framing — as a repository of primary sources that mainstream Huxley scholarship has largely ignored. Julian Huxley has received the most sustained transhumanist-historical attention of the three, particularly since 2015, with Dunér providing the most comprehensive treatment of his role in the movement's origins.

What this chapter contributes is distinct from, rather than in opposition to, existing scholarship. Dunér's account is centrally concerned with Julian Huxley as an origin point for transhumanism, tracing the intellectual genealogy that runs through his evolutionary humanism. This chapter's concern is different: Julian Huxley is not its object but its instrument, a lens through which to trace how rationalism was defined, contested, and transformed across a specific historical moment. The same is true of Stapledon and Aldous Huxley — each is examined not as an isolated figure whose ideas matter in themselves, but as a participant in a proto-transhumanist discourse shaped by the shared pressures of the interwar crisis. What this framework reveals that

existing scholarship does not is threefold. First, it reads these three thinkers as active interlocutors in a contentious and evolving argument, often in direct conversation with one another and with the figures of Chapter 1, rather than as precursors to be catalogued. Second, it recovers Stapledon as an internal critic of proto-transhumanist rationalism — a voice whose philosophical and spiritual dissent has been systematically excluded from transhumanist histories through the same selective appropriation that erased Russell. Third, it argues that the hierarchies embedded in Wellsian rationalism were abandoned by Stapledon, but naturalized by the Huxleys, encoded into the architecture of *Brave New World* and institutionalized in Julian Huxley's *UNESCO* in ways that made them harder to see. Together, these three figures do not represent a progression toward modern transhumanism but a fracturing of it: a moment when the tradition's unresolved tensions were forced into the open by depression, fascism, and the looming specter of another world war.

Julian and Aldous Huxley grew up steeped in the elite English world of science and social planning. Their grandfather, T.H. Huxley—known as "Darwin's bulldog"—had become world-famous as the most prominent advocate for Darwin's theory of evolution. For both brothers, entry into Oxford and high academia was not an impressive achievement so much as an expected one. Julian moved easily among the most eminent thinkers of his generation; as an Oxford undergraduate, he was already dining regularly with figures such as J.B.S. Haldane, who would become a lifelong intellectual peer.⁷⁹ Julian Huxley's most significant intellectual relationship outside his family was with H.G. Wells. The two met in 1926 and by 1927 had begun collaborating on *The Science of Life*, an ambitious survey of biology written for a general audience. They would remain lifelong friends and colleagues, exchanging ideas on evolution,

⁷⁹ Huxley, *Memories*, 65.

social planning, and humanity's future throughout the 1930s and beyond.⁸⁰ Julian attributed the connection partly to his grandfather's legacy—Wells had studied under T.H. Huxley at the Normal School of Science and retained a deep admiration for him throughout his life.⁸¹

Aldous Huxley moved in many of the same circles as his brother. He maintained a direct connection to Haldane Julian Huxley, and Wells himself. Both brothers, Wells, and Stapledon belonged to the Federation of Progressive Societies and Individuals, the *Samurai* organizations inspired by *A Modern Utopia*, and the Next Five Years Group, a 1935 organization that having lost faith in the parliamentary system, advocated for technocratic planning as the engine of social reform. Aldous Huxley, Stapledon, and H.G. Wells were all members of the Fabian society, and.⁸² The overlapping memberships were not incidental; they reflect a shared intellectual world in which proto transhumanists were in constant correspondence, continuously shaping and refining one another's ideas.⁸³

Notably, throughout Aldous Huxley's section, I rely heavily on David Bradshaw's (1955-2016) work *The Hidden Huxley*. David Bradshaw was an accomplished Professor of English literature, spent most of his career at Oxford University, and has been called "one of the greatest scholars of modernism," a field of literature Aldous Huxley is closely associated with.⁸⁴ Bradshaw, like the vast majority if not all of biographers, has some level of bias towards Huxley in his writing, occasionally interpreting the texts and times in an excusatory way for Huxley's more poorly-aged opinions. Despite this, Bradshaw's historical work is thorough. Bradshaw

⁸⁰ Huxley, *Memories*, 154–78.

⁸¹ Huxley, *Memories*, 155.

⁸² David Bradshaw, *The Hidden Huxley: Contempt and Compassion for the Masses* (Faber and Faber, 1994); Julian Huxley, *Memories II* (Harper & Row, 1973).

⁸³ Bradshaw, *Hidden Huxley*, 20–42.

⁸⁴ Barbara Cooke, "Professor David Bradshaw, 1955–2016," *Waugh and Words* (blog), September 2016, <https://staffblogs.le.ac.uk/waughandwords/2016/09/15/professor-david-bradshaw-1955-2016/>.

heavily utilized and incorporated primary sources within *The Hidden Huxley*; he also drew on other historians biographies of close figures, providing an intext cross-reference for his sources. The use of Bradshaw's work is primarily as a repository of primary sources with provided historical context.

Stapledon lacked the intellectual-aristocratic background of the Huxley's. Rather, he came from a middle-class background, pursued a philosophy degree and became a lecturer and writer of philosophy. Stapledon as a leftist intellectual ended up in many of the same groups as Wells and the Huxley's. He was a founding member of the H.G. Wells society and a member of the Fabian society.⁸⁵ Ultimately though, Stapledon differed on many counts from the Huxleys and Wells. Whereas Wells was openly against any form of religion, Stapledon sympathized with spiritual tendencies and merely rejected the organized religions of his day. He was also intensely egalitarian and believed in the capability of the vast majority of the populace to become educated and capable, to the ire of the elitist Aldous Huxley.⁸⁶ Despite this, Stapledon had immense respect for H.G. Wells, telling him that he did not write an outright acknowledgement for Wells' influence on *Last and First Men* because "A man does not record a debt to the air he breathes."⁸⁷ Wells ultimately lost interest in Stapledon as a divide grew between the two men; Stapledon was interested in "agnostic mysticism," while Wells described himself in contrast to Stapledon as "more positivist and finite than you."⁸⁸

⁸⁵ Robert Crossley, *Olaf Stapledon: Speaking for the Future* (Syracuse University Press, 1994), 38, 76, 211–12; Smith, *H.G. Wells*, 332–33.

⁸⁶ Bradshaw, *Hidden Huxley*, 40.

⁸⁷ Patrick Parrinder, *The Earth Is My Footstool: Wells, Stapledon and the Idea of the Post-Human*, January 2012.

⁸⁸ Parrinder, *The Earth Is My Footstool*. This is also where Stapledon refers to Wells as an unknowable and grand "Sphinx."

This chapter traces how the ideological differences within early transhumanist discourse deepened as the interwar crisis—defined by depression, fascism, and the looming specter of another world war—forced these thinkers to grapple with the political and existential crises of their science. Where Haldane and Bernal had written with relatively little doubt about the transformative power of science, the thinkers of this chapter faced those ambitions head-on: Aldous Huxley naturalized a hierarchical worldview similar to Wellsian frameworks into the biological architecture of *Brave New World*, one of the most widely read novels of the twentieth century. The dominant scholarly tradition, represented most influentially by Peter Firchow's reading of the novel as primarily satirical, has situated *Brave New World* as a cautionary tale directed outward at authoritarian excess. This chapter follows Joanne Woiak's argument that the novel is better understood alongside Huxley's nonfiction as a serious engagement with eugenics and social reform, and extends that reading into the specifically proto-transhumanist context Woiak does not address.

Stapledon was an outsider in both origin and writing, born middle class and educated in philosophy rather than hard science. Stapledon embedded his critique of rationalist elitism into a cosmic narrative that imagined global war and authoritarian collapse as the inevitable consequences of rationalism untampered by spirituality and egalitarianism. Stapledon is omitted entirely from Bohan and Bostrom's work and left as a literal footnote in Dunér's work. Julian Huxley attempted the first step of creating the scientific world government his predecessors dreamt of. The scale of this attempt dwarfs anything in chapter one: UNESCO was not a novel or an essay but an institution, the scientific and cultural body of the United Nations itself, carrying assumptions about science, evolution, and human hierarchy into the postwar world order. The war changed what was possible and what was acceptable: eugenics was tainted, and overt elitism

was politically toxic. Julian Huxley's UNESCO stands at the breaking point of two worlds, mixing ideas throughout early transhumanist discourse with the new post-war future. These three responses, ranging from dystopian reckoning to institutional compromise to cosmic hope reveal a discourse far more ideologically fractured than transhumanist histories have acknowledged.

Section II: Stapledon's *Last and First Men*

What follows is not simply an analysis of these authors' ideas, but an exercise in recovery. Robert Chartier's concept of selective appropriation holds that readers and communities reshape texts to fit their own ideological needs, absorbing what is useful and discarding the rest.⁸⁹ The board game Monopoly offers a telling example: originally designed by Elizabeth Magie as a critique of land ownership and capitalist monopoly, it was selectively appropriated into the very celebration of capitalism it was meant to indict.⁹⁰ In transhumanist history, Olaf Stapledon has suffered a similar fate to Magie. Nick Bostrom recognizes Stapledon as popularizing ideas of human future evolution and his creative speculations about humanities future in space. Elise Bohan's detailed work, *A History of Transhumanism* does not mention Stapledon once. Nor does Alexander Thomas's *A Brief History of Transhumanism and its Critics*. Stapledon has been relegated to his speculative technology ideas. Ironically, his most famous idea, the Dyson sphere, was mistakenly named after another author, Freeman Dyson.⁹¹

This reduction of his philosophical critiques, the main substance of his writings, is critical. Transhumanism in the modern day shields itself from philosophical argument, imagining itself as fundamentally improving the human condition in scientific ways, and therefore

⁸⁹ Roger Chartier, *Cultural History: Between Practices and Representations*, trans. Lydia Cochrane (Cornell University Press, 1988), 5-40.

⁹⁰ "Monopoly Go Revenue and Usage Statistics (2025)," *Business of Apps*, n.d., accessed November 23, 2025, <https://www.businessofapps.com/data/monopoly-go-statistics/>; "Monopoly Was Invented to Demonstrate the Evils of Capitalism," July 28, 2017, <https://www.bbc.com/worklife/article/20170728-monopoly-was-invented-to-demonstrate-the-evils-of-capitalism>.

⁹¹ Dyson, *Disturbing the Universe* (1979) Harper & Row.

operating outside the bounds of spiritualism and philosophy.⁹² Spiritualism and philosophy are precisely the fields Stapledon was working in. Bohan's complete omission of Stapledon is the natural consequence of a historiographical method that only recognizes transhumanism when it arrives in a scientific shell. Modern philosophical and spiritual critiques of transhumanism are illegitimately weakened by this inaccurate historization. This section attempts to restore what has been left out, in order to reenvision transhumanist history belonging to anyone who may be interested in transhumanism; atheists, the organized religious, LGBTQ+, agnostics, or countless others. This chapter asserts a different Stapledon: one of the earliest internal critics of transhumanist thought, dissenting from his peers on rationalism, elitism, and gender in ways that have gone largely unexamined.



Figure 5. William Olaf Stapledon.
 Photograph. National Portrait Gallery,
 London.
<https://www.npg.org.uk/collections/search/person/mp48912/william-olaf-stapledon>.

⁹² This self-positioning as a scientific rather than philosophical project is visible throughout transhumanist self-description. See Nick Bostrom, "Transhumanist Values," *Journal of Philosophical Research* 30 (2005): 3; More, "The Philosophy of Transhumanism," *Humanity+*, <https://www.humanityplus.org/philosophy-of-transhumanism>. The historiographical consequence of this posture is Bostrom's own *History of Transhumanist Thought*, which excludes Russell and Stapledon. Roy Jackson's *Muslim or Supermuslim* and Levin's *Posthuman Bliss?* identify this trend. Roy Jackson, *Muslim and Supermuslim: The Quest for the Perfect Being and Beyond* (Springer International Publishing, 2020), 163–74, <https://doi.org/10.1007/978-3-030-37093-0>.

Stapledon's novel *Last and First Men* traces humanity's future across two billion years, beginning in the early 2000s on the brink of war. Again and again, humanity destroys itself and reemerges—sometimes recognizably human, sometimes radically transformed, at one point devolving into small, bunny-like rodents scratching out an existence on Neptune. Each successive species serves Stapledon as a philosophical thought experiment, a way of testing different arrangements of rationality, spirituality, elitism, and community against the long arc of human possibility. It is this philosophical architecture, rather than the novel's speculative technology, that makes it essential to a history of transhumanist thought.

Early in the novel, Socrates and Jesus are presented as equals; both are "essential" to the human spirit, representing the twin peaks of Western civilization. Socrates embodies "glorified unbiased thinking" and "dispassionate intelligence," while Jesus represents love, unselfishness, and what Stapledon calls "self-obliviousness:" traits that are decidedly non-rationalist and non-elitist. From the outset, rationalism is posited as vital but incomplete.⁹³ Stapledon nonetheless takes it seriously: the earlier species of humanity that evolve within the novel all follow rationalism to great ends.

In one striking example, a subspecies splits into two camps—one led by a charismatic leader, one by a rationalist biologist—and Stapledon traces their diverging evolutionary fates across millions of years. The biologist's descendants shed what Stapledon calls the "primitive relics" that had hampered earlier humanity; even their children were "instinctively inclined toward a genuinely aesthetic interest in their world... and also toward scientific inquiry and

⁹³ Olaf Stapledon, "Story of the Near and Far Future: *Last and First Men*," Penguin Books, 1937. 10-20.

generalization."⁹⁴ The charismatic but irrational leader's group, by contrast, had devolved into something barely recognizable: "more baboon-like than human," enslaved by monkeys, their intelligence sacrificed entirely to an overdeveloped sense of smell, their social hierarchy organized around disease and fear rather than reason. Stapledon's point was blunt—irrationalism, left to compound across millennia, does not merely stagnate but catastrophically regresses.⁹⁵

This is not simply a celebration of rationalism. Stapledon is careful to show that the biologist's group succeeds not because intellect alone triumphs, but because their rationalism is paired with aesthetic sensibility—the offspring are drawn to beauty *and* science, in equal measure. This pairing is deliberate and central to Stapledon's broader argument: rationalism without an emotional or spiritual counterweight is, for him, a ceiling rather than a summit. He admires the biologist's descendants, but they are not his ideal. They are a necessary step and a vast improvement over charismatic irrationalism, but the novel's entire arc is building toward something rationalism alone cannot reach. Stapledon plants this seed early, and when his later species transcend pure intellect entirely, it reads not as a rejection of science but as its fulfillment.

The ceiling that rationalism hits in the biologist's descendants becomes a structural feature of the novel. Stapledon's Fourth Men, likely inspired by J.D. Bernal's contemporaneous speculations about disembodied intellect, are vast brains suspended in vats, their biological bodies replaced entirely by mechanical extensions. For tens of thousands of years, they dominate the globe, advancing science and intellect beyond anything prior species achieved. And then they stall. Introspection reveals the cause: without emotion, spirituality, or mortality, existence has

⁹⁴ Stapledon, *Story of the Near and Far Future*, 115-119.

⁹⁵ Stapledon, *Story of the Near and Far Future*, 121

become static. Purpose requires more than intellect can supply. Stapledon's response is telling: the Fourth Men do not push harder into science, but instead engineer the Fifth Men, who retain enormous cognitive capacity while reintroducing emotion, relationships, and crucially, death.⁹⁶ Buried in this storytelling is one of Stapledon's most original and unacknowledged contributions to transhumanist thought: an early, direct challenge to the idea of immortality as an unqualified good. Where most of his peers either ignored the question or treated eternal life as an obvious goal worth striving for, Stapledon presents it as a trap, a condition that drains existence of dynamism and purpose rather than fulfilling it.

If the Fourth Men represent rationalism's ceiling, the Last Men represent Stapledon's answer to what lies beyond it. They are not the novel's most scientifically capable species—the brain-vat humans rival or exceed them in pure intellect—but they are its most fully realized. What elevates them is the integration of scientific knowledge with emotional depth and an unwavering commitment to the welfare of all conscious life. This shift is reflected in the novel's structure itself: the final chapters largely abandon the biological and technological progress that drove earlier sections, turning instead to the spiritual. The Last Men practice something resembling religion, although Stapledon resists the term as inadequate, calling it merely the closest word available to a lesser species, and maintain ritual practices, including communal eating, long after food has ceased to be nutritionally beneficial. Even as they face the extinction of their solar system, they find spiritual solace in what they understand as their own spiritual eternity by sending out the seed of life to another solar system. This is the culmination of Stapledon's argument, the point toward which the entire arc of human evolution has been building.⁹⁷

⁹⁶ Stapledon, *Story of the Near and Far Future*, 170-200.

⁹⁷ Stapledon, *Story of the Near and Far Future*, 257-286.

The Last Men use science not as an end in itself but as an instrument of spiritual and communal deepening. Most strikingly, Stapledon imagines them reaching backward through time to influence Jesus, Socrates, and Gautama, the very figures held up as civilization's twin pillars in the novel's opening pages. The loop is deliberate: the greatest species humanity will ever produce turns out to be the secret hand behind the greatest humans who ever lived, and what they chose to cultivate in those figures was not just rationalism but also love, wisdom, and self-transcendence. In another instance, biological modification enables species-wide mind-melding, allowing the Last Men to strive toward what Stapledon calls the "Flower of All Things... All-Knowing... All-Admiring." The capitalization is not accidental, but rather consciously echoes the Western theological tradition of rendering the divine in uppercase, recasting the omniscient not as an external God but as a collective human aspiration realized through science. In doing so, Stapledon charts a path; rather than discarding religion as irrational superstition, he reimagines it as the destination that science, properly directed, might actually reach.⁹⁸

What emerges in the Last Men is a self-reinforcing cycle—spirituality motivates science, science deepens spiritual capacity, which in turn demands greater science. Their final act crystallizes this perfectly: the seeding of life across the dying solar system, an action Stapledon himself notes is not strictly rational, but instead driven by love and a sense of cosmic obligation. The goal he assigns them, to "seek then to know and love every past mind," places love, not logic, as the animating force behind their most ambitious scientific undertaking.⁹⁹ Rationalism has been demoted from sovereign ideology to servant tool in the hands of something larger. This

⁹⁸ Stapledon, *Story of the Near and Far Future*, 286-294.

⁹⁹ Stapledon, *Story of the Near and Far Future*, 293.

is Stapledon's radical conclusion: that the arc of human evolution bends not toward pure reason but toward wisdom, and that wisdom requires feeling as much as thought.

The novel's final passage makes no mention of rationalism or science. Stapledon closes instead with beauty, music, and cosmic belonging—"Man himself, at the very least, is music, a brave theme that makes music also of its vast accompaniment, its matrix of storms and stars. Man himself in his degree is eternally a beauty in the eternal form of things."¹⁰⁰ It is a conclusion baffling to a rationalist. A call simply to recognize in humanity something already and eternally worthy of wonder, a spiritual humanism.¹⁰¹

Where Wells viewed women as a threat to his rationalist engineers and Aldous Huxley would strip them of voice and agency entirely, Stapledon stands apart. He avoided the condemnation of feminine attributes so common among his peers, and gave female characters in his fiction a degree of agency largely absent from the work of Wells or either Huxley brother. His prose habits reflect this too—Stapledon routinely wrote "he or she" at a time when his many of his contemporaries used masculine pronouns universally. His egalitarianism extended outward as well; his *Circular Letter to All the Peoples of the Earth* addressed humanity without exclusion. These are small signals, but they are consistent.¹⁰²

The sharpest illustration of this Stapledon-peers divide comes from an essay written by Aldous Huxley, for other members of the Federation of Progressive Societies and Individuals, in which Huxley critiques a now-lost Stapledon essay on mass education. Huxley's tone is withering: Stapledon, Huxley writes, "tries to escape from this rather depressing conclusion by

¹⁰⁰ Stapledon, "Story of the Near and Far Future." 294

¹⁰¹ Stapledon, "Story of the Near and Far Future."

¹⁰² "Circular Letter to All the Peoples of the Earth with Various Drafts of the List of Persons to Whom It Was to Be Sent," University of Liverpool Digital Heritage Lab, accessed November 23, 2025, <https://digitalheritagelab.liverpool.ac.uk/Documents/Detail/circular-letter-to-all-the-peoples-of-the-earth-with-various-drafts-of-the-list-of-persons-to-whom-it-was-to-be-sent/81229>.

the simple process of loudly declaring that people ought to be different from what they actually are." When Stapledon argues that individuals should be taught to think critically, imaginatively, and independently, Huxley's patience runs out entirely—"He might also, while he was about it, learn to be both Mozart and Sir Isaac Newton").¹⁰³ The sarcasm is revealing. Where Stapledon saw education as a genuine path to widespread human development, Huxley saw such optimism as naive to the point of absurdity. The letter lays bare, more vividly than any theoretical statement could, the fundamental divide between the two men.

The Circular Letter crystallizes what the novel demonstrates at length: Stapledon's anti-imperialism and radical egalitarianism were not incidental to his thought but foundational to it. "No people is so much nobler and wiser than others that it should strive to master the world or rule any other people"—a statement that placed him in direct opposition to the civilizational hierarchies that Wells and the Huxleys largely took for granted.¹⁰⁴ Within the landscape of transhumanist thought, Stapledon was an outlier: skeptical of immortality, critical of pure rationalism, sympathetic to spirituality, and committed to a vision of human advancement that included rather than managed the majority of humanity. And yet he remains undeniably transhumanist—the ascent beyond current human limitations is as central to *Last and First Men* as it is to anything Wells or Julian Huxley wrote. What distinguishes Stapledon is not the destination but the terms: who gets to come, what gets left behind, and what, beyond reason alone, might guide the journey. Still asking those questions, we turn now to Aldous Huxley.

¹⁰³ Bradshaw, *Hidden Huxley*, 40.

¹⁰⁴ "Circular Letter to All the Peoples of the Earth with Various Drafts of the List of Persons to Whom It Was to Be Sent," University of Liverpool Digital Heritage Lab, accessed November 23, 2025, <https://digitalheritagelab.liverpool.ac.uk/Documents/Detail/circular-letter-to-all-the-peoples-of-the-earth-with-various-drafts-of-the-list-of-persons-to-whom-it-was-to-be-sent/81229>.

Section III: Aldous Huxley and Hierarchy

Of the figures examined in this chapter, Aldous Huxley lands closest to the Wellsian model of rationalist hierarchy. He shared Wells's conviction that the intellectual elite were not merely useful but indispensable—that civilization's fate rested in the hands of a scientifically minded few, and that democracy, by empowering the many, was actively dangerous. He embraced Wells's "Samurai" concept without reservation, writing that "we may either persist in our present course, which is disastrous, or we must abandon democracy and allow ourselves to be ruled dictatorially by men who will compel us to do and suffer what a rational foresight demands." Science, in this framing, was not merely a mode of inquiry but a justification for power, high enough in Huxley's worldview, as he put it, to topple both democracy and equality.¹⁰⁵



Figure 6. Aldous Huxley, Riviera, Côte d'Azur. Photograph.
Sourced
from https://cdm01fbn.cloudimg.io/v7/images_base/image_uploader/photos/83/original/aldous-huxley-riviera-cote-d-azur.jpg?ua=1767168725&p=larger

¹⁰⁵ Bradshaw, *Hidden Huxley*, 22.

Huxley's letter to Stapledon reveals not just condescension but a fully formed worldview. Huxley was not merely skeptical of mass education as a practical matter—he considered the intellectual elevation of ordinary people to be, as he wrote, "almost infinitely improbable," not within centuries or even millennia. History, he argued, showed that "most people will accept reason only in small doses... only on irrational grounds, generally of a religious nature."¹⁰⁶ This was a fact of Huxley's reality: the capacity for genuine rational thought was, for Huxley, a rare biological endowment, not a latent potential waiting to be unlocked by better schooling. The elite were few, their capabilities fixed, and their disproportionate influence utterly correct. It is this belief, that the scientifically minded few are the only ones whose judgment truly matters, that saturates *Brave New World*, not as a dystopian warning but the assumption of hierarchy as a given bleeding through every page.

Section IV: Not So Satirical: *Brave New World*

Brave New World depicts a future where laboratories decant, engineer, and condition human beings before birth to fit predetermined social roles, directly inspired by Haldane's ectogenesis proposals. At the top sit the Alphas and Beta-Pluses, intellectually capable and autonomous. Below them are increasingly stunted castes, their development deliberately arrested by alcohol or oxygen deprivation in the fetal stage, destined for menial labor and incapable of questioning their lot. The World State maintains society through pleasure, Soma—a powerful pleasure drug—and consumerism. The novel follows Bernard Marx, a misfit Alpha, and Lenina, a conventional Beta, as they encounter John "the Savage," a man raised outside the

¹⁰⁶ Bradshaw, *Hidden Huxley*, 40.

system who has read Shakespeare and retained something resembling genuine feeling. The collision between John's raw humanity and the World State's frictionless order forms the novel's central tension, and the book reaches its climax in a philosophical conversation between John and one of seven leaders of the World State, Mustapha Mond.

Aldous Huxley wrote *Brave New World* in 1931, at the height of the interwar period—a moment defined by the wreckage of one world war and the gathering shadow of another. Fascism in Europe was born after WWI and was surging in popularity in the thirties. Communism had produced a powerful state in the USSR, and the future of the world seemed more and more uncertain. Huxley was a member of a number of societies with competing visions of the future: some, like the Federation of Progressive Individuals, hoped for democracy, while others, like the H.G. Wells Society, favored technocratic states. We can never know Huxley's true intentions.

The initial reception of *Brave New World* was more complicated than its modern praise suggests. The reactions were scattered, but almost all reviewers interpreted it as a satire of the Utopias the early 1900s had been producing—a number of them outright named *Anticipations* or Wells. Similarly, the vast majority of reviewers reacted with a gruesome horror to Huxley's world. Yet there was no consensus that the book was well executed; many found it boring or self-serving.¹⁰⁷ When read in late twentieth and twenty-first century American and English classrooms, *Brave New World* is interpreted through the lens of democracy, and the idea of an authoritarian government is fundamentally negative. But in the chaotic interwar period, where

¹⁰⁷ Butcher, Fanny. "Aldous Huxley Pokes Fun at Model T World." *Chicago Daily Tribune*, February 8, 1932. C. D. M. "Huxley Turns Propagandist." *New York Herald Tribune*, February 7, 1932. Gannett, Lewis. "Books and Things." *New York Herald Tribune*, February 4, 1932. K., K. "Brave New World." *Country Life* 71 (February 13, 1932): 182. "Mr. Huxley's Horrible World." *The Bystander* 113 (February 17, 1932): 332.

the future was uncertain and no one had the answers, depictions of both failed democracies and failed authoritarian states were not uncommon, and Huxley himself straddled both worlds. Despite this complicated initial reception, *Brave New World* ranked fifth on Modern Library's list of the hundred best novels of the twentieth century, and its vision of a scientifically managed dystopia has shaped popular and intellectual culture far beyond literary circles.¹⁰⁸

This provides a different lens to understand *Brave New World*, obfuscating its role in early political transhumanist discourse. As such, this analysis disregards the unanswerable questions about intention in *Brave New World*. Instead, I examine who Huxley grants a level of integrity and dignity beyond what is natural in dystopia, and who Huxley casts as belonging in a dystopia. This is analyzed to expose internal hierarchies within this essential and famous piece of early transhumanist thought. This analysis builds on Joanne Woiak, Associate Professor of Disability Studies at the University of Washington, whose work on *Brave New World* argues the novel is best understood alongside Huxley's nonfiction as a serious engagement with eugenics and social reform rather than pure satire. This chapter differs from Woiak in the lenses of rationalism, and transhumanism, as well as the exact primary sources consulted. This approach departs from Peter Firchow, who served as Professor of English and Comparative Literature at the University of Minnesota and the leading scholarly authority on Huxley's fiction, whose influential reading situates the novel primarily as satire. I draw additionally on Bradshaw's *The Hidden Huxley* to strengthen interpretations of the novel through Huxley's essays and letters.¹⁰⁹

¹⁰⁸ "The Greatest Books: The Modern Library | 100 Best Novels from Modern Library," accessed October 14, 2025, <https://thegreatestbooks.org/lists/2>.

¹⁰⁹ Joanne Woiak, "Designing a Brave New World: Eugenics, Politics, and Fiction," *The Public Historian* 29, no. 3 (2007): 105–129. Peter Firchow, *The End of Utopia: A Study of Aldous Huxley's Brave New World* (Bucknell University Press, 1984). Bradshaw, *The Hidden Huxley*.

Section V: Hierarchy In *Brave New World*

The architecture of *Brave New World* is unambiguously elitist—and notably, it is an elitism rooted in biology rather than circumstance. J.B.S. Haldane had already imagined something close to this in his 1924 essay *Daedalus*, where he speculated enthusiastically about ectogenesis, the artificial gestation of human beings outside the womb, as a tool for improving the species. Huxley utilized the same technology and revealed its logical endpoint: a world where social destiny is engineered before birth, where those assigned to menial labor have their development chemically arrested in the fetal stage, and where the question of who gets to think, feel, and speak is settled before a single breath is drawn.

The novel's moral conversations—about freedom, truth, and what makes life worth living—are conducted exclusively among Alphas, with Mustapha Mond and John the Savage at the apex of the conversation, and Alpha Bernard Marx barely able to comprehend, his limitations rumored to stem from alcohol accidentally introduced into his fetal environment. No one below Alpha rank is granted the intelligence to participate. This is not incidental. In his 1934 essay "What Is Happening to Our Population?" Huxley wrote plainly that feeble-mindedness was heritable, that the majority of the intellectually limited were born of similarly limited parents, and that the increasing birth rate of the lower class relative to the high class was a social problem demanding intervention.¹¹⁰ His solution was to prevent "the sub-normal from having any families at all."¹¹¹ *Brave New World* does not satirize this belief, but rather builds a world on top of it.

¹¹⁰ Aldous Huxley, "What Is Happening to Our Population?," in *Debating Malthus*, (University of Washington Press, 1934), 2–9, <https://www.degruyterbrill.com/document/doi/10.1515/9780295749914-032/html>.

¹¹¹ Huxley, "What Is Happening to Our Population?," 5.

This hierarchy extended beyond politics into culture, and it is here that Huxley's worldview most resembles Wells. David Bradshaw, in *The Hidden Huxley: Contempt and Compassion for the Masses*, argues a direct line from Huxley to Nietzsche on this point. Where Nietzsche argued that culture and art required the enslavement of the majority, that "the misery of toiling men must still increase in order to make the production of the world of art possible to a small number of Olympian men," Huxley made the same case in more palatable language, lamenting that democracy destroyed "something grand and splendid," replacing aristocratic culture with something "sordid and small."¹¹² His ideal, stated plainly in his own words, was "a material democracy controlled by an aristocracy of intellect," a society stratified not by birth but by cognitive and cultural capability, which in practice amounted to much the same thing.¹¹³ Considering Huxley's direct line to Wells, it is difficult to say if this is Wellsian inspired, if Bradshaw is right and that it is Nietzschean, or something else, but the throughway of aristocratic-forward thought between the three writers is undeniable. The gulf between this aristocracy and everyone else was, for Huxley, not merely growing but unbridgeable: he wrote to H.L. Mencken, an American journalist and peer that within twenty years, "a man of science or a serious artist will need an interpreter in order to talk to a cinema proprietor or a member of his audience."¹¹⁴

The masses were not merely less rational than the elite; they were becoming a different kind of creature entirely, receding from the possibility of genuine civilization. In this way, Huxley's framework escalates Wellsian style hierarchy to an extreme: the elite are not just the

¹¹² Bradshaw, *Hidden Huxley*, 12–19. I could not earnestly cite anything except Bradshaw for this. Bradshaw tracked down Huxley's anonymous review on a Mencken article through cross referencing with other magazines and letters.

¹¹³ Aldous Huxley, "The Outlook for American Culture," *Harper's Magazine*, August 1927.

¹¹⁴ Bradshaw, *Hidden Huxley*, 19–20.

most capable, they are the only ones whose inner lives, judgments, and cultural productions actually matter. In *Brave New World*, this logic reaches its endpoint. Mustapha Mond, one of the World State's seven Controllers, explains the system's logic with chilling clarity: each person lives, he says, inside a bottle—but "if we happen to be Alphas [the high caste], our bottles are, relatively speaking, enormous," while Epsilon [the low caste] conditioning has "laid down rails along which he's got to run. He can't help himself; he's foredoomed."¹¹⁵ This is not pure satire or portrayed as fundamentally wrong. Huxley does leave room for John to protest, but crucially, John never reaches any philosophical triumph over Mustapha: The two ideologies end in a stalemate. Whether John or Mustapha—individual freedom or stratified society—is right is entirely left to the reader to decide.¹¹⁶

Where Huxley diverges from Wells is on the question of emotion. For Wells, feeling was essentially a liability—irrationalism in disguise, to be subordinated entirely to rational planning. Huxley granted emotion more room but circumscribed it carefully: feeling was real and valuable, but only in those capable of experiencing it with sufficient depth and refinement, which is to say, only in the elite. For everyone else, emotion was not a path to truth but a vector for manipulation. This is made explicit in *Brave New World's* dystopian logic: when a propagandist declares that "moral education ought never, in any circumstances, to be rational," Huxley frames it as one of the World State's most chilling inversions.¹¹⁷ The joke lands if the reader shares Huxley's premise—that rationality is the only legitimate foundation for morality, and that a society severing that connection has lost something essential. Emotion, then, is not the enemy of Huxley's rationalism but its dependent: valuable when disciplined by reason, catastrophic when

¹¹⁵ Aldous Huxley, *Brave New World* (Chatto & Windus, 1931), 96.

¹¹⁶ Huxley, *Brave New World*, 94-104.

¹¹⁷ Huxley, *Brave New World*, 11.

untethered from it. Reason, moreover, was a faculty Huxley believed to be unequally distributed across humanity in ways that were biological, fixed, and consequential.

Huxley's fear of unmediated emotion was not abstract—he saw it as a direct political threat. Fascism, he argued, would come to England not through reasoned argument but through "the logic of emotions," winning entirely "on the emotional plane."¹¹⁸ The masses, ungoverned by rational discipline, were permanently susceptible to demagogues who knew how to feel rather than think. Huxley even suggests that the fight against fascism would require an anti-fascist religion and governmental system that provided "all the amusements they need to make life seem bearable."¹¹⁹ In an eerily similar fashion, Mustapha Mond likens *Soma*, a happiness drug, "Christianity without tears" while explaining how order in the society is maintained, and the government of *Brave New World* requires amusement and celebration time in its citizens schedule.¹²⁰ Furthermore, Huxley did not believe all people were equally at the mercy of their emotions. Those trained in biology, he argued, had developed a unique capacity to think scientifically about what was "emotionally near"—their own bodies, their own vital processes—and this discipline, once acquired, extended to everything else.¹²¹ Biology was, in his words, "the least compatible with superstition" of all the sciences.¹²²

This reverence for biology was biographical. Aldous Huxley had grown up in the shadow of T.H. Huxley, whose defense of Darwin had made biology the standard-bearer of rational thought in English intellectual life, and alongside Julian Huxley, who had built his entire career

¹¹⁸ Aldous Huxley, *Between the Wars: Essays and Letters*, ed. David Bradshaw (I. R. Dee, 1994), 140–41. This is not an interpretative piece by Bradshaw, but rather a collection of Huxley's essays and letters. The specific essay is *Fascism in England*, which was never formally published.

¹¹⁹ Huxley, *Between the Wars*, 140.

¹²⁰ Huxley, *Brave New World*, 90–102.

¹²¹ Huxley, *Between the Wars*, 134.

¹²² Huxley, *Between the Wars*, 134.

on the premise that biological science was humanity's most powerful tool for self-understanding and self-improvement. For Aldous Huxley, biology was not merely a discipline but a worldview, one he had absorbed from childhood. The biologically educated were, in effect, the only people capable of handling their own inner lives without becoming dangerous. In Huxley's time, women were largely excluded from the biological sciences, and those few who did gain entry were routinely dismissed and denigrated. To locate rational emotional mastery exclusively in the biologically trained was, structurally, to locate it exclusively in men. This was not an accidental byproduct of Huxley's elitism but its logical consequence—and nowhere is that consequence more visible than in how he populated *Brave New World* with women.

Section VI: Gender and *Brave New World*

Huxley's depiction of women in *Brave New World* reflect they inhabited a world organized around a gendered hierarchy. Lenina, the novel's most prominent woman, is defined almost entirely by cheerful compliance; her most revealing and sinister moment comes when she declares "I don't understand anything," not with distress but with satisfaction, "determined to preserve her incomprehension intact."¹²³ Fanny, her friend, exists almost solely to gossip and admire consumer goods. Both women are incapable of the moral conversations that define the novel's male characters, and neither shows any desire to be otherwise.¹²⁴ As Deanna Madden observes, the novel sends the consistent message that women are "less intelligent, less capable of seeing beyond their own immediate physical comforts, and less likely to make heroic gesture or defy status quo."¹²⁵ Madden's observation points toward but did not outright state that this was

¹²³ Huxley, *Brave New World*, 40.

¹²⁴ Huxley, *Brave New World*, 23.

¹²⁵ Deanna Madden, "Clockwork Orange," in *Misogyny in Literature: An Essay Collection*, ed. Katherine Anne Ackley, Garland Library, 1992.

not merely faulty characterization within a novel, but ideology embodied. Huxley is not depicting a dystopia that happens to oppress women; he is writing women as dystopia's natural inhabitants, a species so thoroughly at home in a world of conditioning and consumption that they require no coercion at all.

Linda, the third woman, takes this furthest—given freedom, she uses it only to crave and consume Soma, an addictive pleasure drug, in ever larger doses until it kills her.¹²⁶ That both Linda and Lenina show no interest in Shakespeare, the novel's marker of genuine interiority and moral seriousness, connects them to Bernard Marx, whose own intellectual limitations are explicitly attributed to fetal alcohol poisoning.¹²⁷ The implication is entirely unexamined by Huxley himself: women's failure to engage with high culture is presented as natural rather than conditioned, a biological given rather than a social product of dystopia. No “Alpha” status women exist in the book, which is never examined. Every positive quality Huxley distributes across his male characters—courage, creativity, autonomy, intellectuality—is withheld from his women entirely, not as critique but as assumption. It is never explained the way all male characters limitations and exceptionalities are described.

In Huxley's work, rationalism is not the dominant force. In fact, it is never specified as a required element for either democracies or authoritarian societies, both suggested to perform better through a state religion. The titular philosophical discussion at the end of *Brave New World* is not argued through figures or facts, but interpretations of Shakespeare and what degree of suffering is worth individual freedom. Instead, the defining trait of Huxley's works is hierarchy, of genetics, of gender, of class, and of fundamental capability. This not only informs

¹²⁶ Huxley, *Brave New World*. 65.

¹²⁷ Huxley, *Brave New World*. 56, 65, 82.

how much influence a person should have in society but taps into a Nietzschean style construction of society as for the ultra-elite. This is Wellsian in nature, if not influence. Despite this, Aldous Huxley is not the Huxley brother that worked closest with H.G. Wells. That title goes to Julian Huxley.

Section VII: Julian Huxley and *UNESCO*.

Julian Huxley's rationalism is best examined through *UNESCO: Its Purpose and Philosophy*, the document he wrote in 1946 to lay the ideological groundwork for the United Nations Educational Scientific and Cultural Organization, of which he became the first Director-General. In the aftermath of World War II, the United Nations became desirable as a tool to prevent another World War. As the grandson of T.H. Huxley and a lifelong collaborator of H.G. Wells, Julian brought to UNESCO a worldview already deeply formed: evolution as humanity's central story, science as its most reliable guide, and a biologically trained elite as its most capable stewards.¹²⁸ The document's reception was disastrous. Written in the immediate aftermath of World War II, when fears of cultural imperialism ran high and eugenics had been permanently tainted by the Nazis' Holocaust, Huxley's vision of a powerful, ideologically unified global UNESCO was refused publication by its earliest committees.¹²⁹ The document today appears prominently on UNESCO's own website, described as a "seminal text."¹³⁰ When brought together with two excerpts from his 1926 *Essays of a Biologist*, these documents collectively show the continuity of his thought across two decades. Dunér's account of Julian Huxley is the

¹²⁸ Julian Huxley, "UNESCO: Its Purpose and Philosophy," UNESCO, 1946, <https://unesco.org.uk/resources/unesco-its-purpose-and-philosophy-julian-huxley-1946>.

¹²⁹ Clark Sorensen, "The Evolutionary Humanism of Julian Huxley and the Creation of UNESCO," *Journal of the History of Ideas* 60, no. 3 (1999): 531–53.

¹³⁰ Huxley, "UNESCO: Its Purpose and Philosophy." See citation 124 for the web page!

most comprehensive available and this chapter does not dispute it. Where Dunér traces Huxley as an origin point for transhumanism through his evolutionary humanism, this section pursues a different question: how Huxley's work continued to reshape transhumanist rationalism, and an analysis of Huxley as a continuation of the spiritual thought of previous thinkers, such as Stapledon or Wells.



Figure 7. Julian Huxley, founder of UNESCO. Photograph. Sourced from https://nl.sott.net/image/s30/610372/full/1_Julian_Huxley_Founder_of_UNE.jpg

The ideological core of *UNESCO* is what Huxley calls "scientific world humanism, global in extent and evolutionary in background."¹³¹ This is a significant departure from the stern rationalism of Wells, and even from the biological determinism of Huxley's own earlier work. Rationalism, Huxley argues, would bias UNESCO just as surely would a focus on spiritualism or economics. Rationalism is too narrow and too easily weaponized in service of the hierarchies humanism is meant to dismantle. Scientific humanism is his attempt to build something greater:

¹³¹ Huxley, "UNESCO" Pages 7-10.

a framework in which science remains the supreme mode of inquiry, while wedded to a genuine commitment to human dignity, equality of opportunity, and the welfare of all peoples regardless of background. In theory at least, this places humanism as the governing principle, with science as its instrument rather than its master. It is the most significant evolution in Huxley's thought relative to his peers, and it marks a real, if incomplete, break from the Wellsian tradition he inherited. How incomplete that break was becomes clear when the details of his humanism are examined more closely.

The limitations of Huxley's humanism become apparent the moment its details are examined. He advocated for increasing genetic variability across humanity, encouraging isolated populations to interbreed with outside communities in the interest of the species. He maintained that all ethnicities were capable of intellectual greatness, but not the same intellectual greatness. Different nationalities, in his view, were genetically predisposed toward different cognitive strengths, one perhaps excelling at mathematics, another at physics. This was humanism on a group-to-group basis rather than a human-to-human one, and it sat uneasily with any genuine commitment to individual equality. It was a movement forward from biological anthropology of the past that tried to assert different groups as more intelligent or superior to other groups, but it was still far from any true egalitarianism. Antithesis to the very humanism he preached still was his advocacy for profiling people into "psycho-physical types" and assigning them to professionally suitable roles accordingly. This was a system he presented as the solution to democracy's central flaw: that it allowed the less intellectually capable as well as the manipulative to acquire political power.¹³² The contradiction at the heart of Huxley's project is stark. He believed in equality of dignity, opportunity and intensely desired to dismantle

¹³² Huxley, "UNESCO" Page 42.

socioeconomic factors he saw as limiting human potential. But he simultaneously believed in fixed genetic differences in capability that only a powerful centralized institution like UNESCO could properly manage. In practice, his humanism utilized the same paternalistic control over individual lives that his brother's elitism did, differing mainly in its tone. Where Aldous Huxley was contemptuous towards the masses, Julian Huxley imagined himself benevolent. The outcome, for those assigned their place by scientific authority, was much the same.

Huxley's scientific humanism was deliberately expansive in scope. Rather than confining UNESCO to the hard sciences, he argued it should encompass social sciences, art, and culture, and should maintain a dedicated philosophy division tasked with the ongoing "quest for a restatement of morality."¹³³ This would be a living, evolving ethical-cultural framework rather than a fixed one. This is where his thought becomes most distinctly his own, and most relevant to the history of transhumanism. Evolution, for Huxley, was not merely a biological process but a cosmic one, extending beyond genetic change into the accumulation of knowledge, the expansion of social organization, and the increasing velocity of human progress. Humanity, he argued, was the "sole heir of evolutionary progress to date:" the universe's most complex product, and therefore its steward. From this he derived something close to a secular theology: mankind had a *destiny*, defined by evolution itself, to realize "the maximum progress in the minimum time."¹³⁴ Science, art, morality, and culture were all instruments of this destiny, which is why all of them fell within UNESCO's proper domain. This line of thought reflected proto-transhumanism in one of its most religious forms—not merely the hope that humanity might improve itself, but the conviction that improvement is written into the structure of the universe,

¹³³ Huxley, "UNESCO" Pages 40-44.

¹³⁴ Huxley, "UNESCO." Page 10.

and that a scientifically guided elite, operating through UNESCO, possessed both the tools and the obligation to accelerate it.¹³⁵ The ghost of Wells's Samurai was still present, but dressed in the language of cosmic purpose rather than technocratic management.

The contradiction is structural rather than incidental. Huxley champions diversity of national cultures as essential to human progress yet concludes *UNESCO* by declaring the organization's single purpose to be "the emergence of a single world culture." If evolutionary humanism is a universal truth, then cultures that have not adopted it are simply behind and drawing them into the framework becomes obligation rather than imperialism. It is Wells's *Samurai* gone global. It is precisely this logic that made the document unpublishable in 1946, when the nations forming UNESCO had just fought a war against any kind of civilizational universalism.¹³⁶

Huxley's relationship with religion tracked the evolution of his broader thought. In his earlier career he had been cast by the religious press as a militant atheist bent on stripping society of its moral foundations, and there was some truth to that characterization. His early essay *Rationalism and the Idea of God* positioned rationalism as the antidote to the chaos and superstition he associated with unchecked religious belief.¹³⁷ But Huxley was not as dismissive of religion as Wells, who regarded it as straightforwardly and unambiguously harmful. Over time Huxley came to see religious feeling, the sense of purpose, moral seriousness, and connection to something larger than oneself, as a genuine human need that science could not simply abolish. Evolutionary humanism was his answer: a framework that preserved the emotional and moral

¹³⁵ Huxley, "UNESCO." Page 7

¹³⁶ Huxley, "UNESCO." Page 57.

¹³⁷ Dunér, *Julian Huxley*, 65; Julian Huxley, *Essays of a Biologist* (Chatto & Windus, 1923), 205–31.

architecture of religion while replacing its dependence on an unknowable God with the knowable trajectory of evolution.¹³⁸ Humanity had a destiny, but it was one science could read and guide. In this way, Julian Huxley's thought fell between that of Wells and Stapledon—less hostile to spirituality than the former, less willing to grant it independence from scientific oversight than the latter. It is an ideology thought out over decades, a characteristically Julian Huxley position, and it is both his most original contribution to proto-transhumanist thought and a clear sign of how far the tradition had traveled from its Wellsian origins. Where Stapledon had imagined humanity reaching toward something he could only call spiritual, and Clarke would later render that reaching as cosmic transcendence, Huxley attempted something more audacious: to build the institution that would steward it. Evolutionary humanism was not just a philosophy but a secular theology, and UNESCO was its church.

Huxley did not fit squarely into the box of any other thinker in his time. He worshipped science as did Haldane, but he did not wage war on religion and instead accepted spirituality as part of humanism. Huxley did not put rationalism above all else, but he still shared a Wellsian view of scientists as the elite capable of deciding humanities morality and future, rather than championing democracy. Huxley did not believe in eugenics and racial superiority in the Wellsian fashion, but he did believe in genetic based differences in capability from person to person and still advocated for a strange type of eugenics and profiling based on “psycho-social types” rather than race. A Wellsian sense of elitism whereby certain individuals who are scientifically minded are fit to lead and those who are of the wrong mindset are not fit to lead still lives on in Huxley's work. What we see in Huxley is the evolution of proto-transhumanist

¹³⁸ Dunér, *Julian Huxley*, 56–67.

thought, moving past the harsh ‘all-science no religion’ of Haldane or the stern rationalism of Wells, to create a blend that emerges as “evolutionary humanism.”

Section VIII: Conclusion

The three figures examined in this chapter inhabited the same intellectual world, read the same books, belonged to the same organizations, and wrestled with the same questions about humanity's future. Both Julian Huxley and Stapledon formed spiritual frameworks around transhumanism. Yet their answers diverged in ways that map onto a single underlying fault line: what role rationalism should play in human advancement, who gets to exercise it, and what it costs to place it at the center of a civilizational vision. Wells had established the template: a gendered, elitist rationalism in which a scientifically trained male elite would manage humanity's ascent, with democracy, emotion, and religion subordinated accordingly. Aldous Huxley inherited this template and articulated an elitism at once biological and permanent. In his framework, women were the dystopia's natural inhabitants rather than its victims, and the outsized role of the biologically educated few left no meaningful room for the majority of humanity to participate in its own future. That he believed himself a humanist made his assumptions harder to see, not easier. Julian Huxley attempted the most deliberate evolution of the tradition, absorbing rationalism into the broader framework of evolutionary humanism and making genuine, if incomplete, commitment to human dignity and cultural equality. But the *Samurai* never fully disappeared from his thought—they simply acquired a global institution and a cosmic mandate.

Stapledon alone broke the mold in a fundamental sense. His rationalism was real but provisional, a steppingstone rather than a destination, and his vision of human advancement was the only one in this chapter that did not quietly require the majority of humanity to be managed, conditioned, or left behind. Transhumanist history has largely forgotten this distinction, instead celebrating these figures together as pioneers of human progress while smoothing over the profound differences in whose progress they actually imagined. This is a form of selective appropriation, and at the crux of this thesis. Recovering those differences is not merely an academic exercise. The questions these men argued about in drawing rooms and philosophy lectures and UNESCO committees persist today: who decides humanity's future, by what values, and for whom. The answers have not yet been settled.

Cosmic Visions and Forgotten Futures, 1968–1977

Section I: The Third Window

In 1968, audiences sitting in darkened cinemas watched something unprecedented: humanity's end. Not through catastrophe or war, but through transcendence. A lone astronaut, spinning through the void beyond Jupiter, dissolved in pulses of alien light and reemerged as something else entirely, and a luminous being of unknown purpose gazed back at the small blue planet he had left behind. Stanley Kubrick's camera offered no explanation. Arthur C. Clarke, who wrote the novel simultaneously with the screenplay, offered only the barest outline: something had guided human evolution from the beginning, and now it was guiding it again, toward a destination no human language could describe. *2001: A Space Odyssey* was not a film that planned a technological future, but instead an epic of human transcendence.

A year later, FM 2030 published *Optimism One*. Where Clarke had written through symbolism and imagery, FM 2030 wrote in fragments, barely formed sentences that he believed would become the communication style of the future. Clarke haunts *Optimism One*—his ideas appear throughout, cited and absorbed—but FM 2030 ran them through a wholly different lens. He was, at his core, a social planner in the tradition of Wells and the Huxleys: imagining the reorganization of governments, economies, and cultures, convinced that technology's greatest promise was liberation—the freeing of individuals from poverty, nationalism, and biological limitation. Whereas Clarke asked what humanity might transcend into, FM 2030 asked what humanity needed to be freed from. These were not opposing transhumanisms but rather two gazes fixed on the same technological horizon, seeing entirely different futures.

This chapter argues that the 1960s and 1970s mark the moment proto-transhumanist discourse escaped its elite British institutional origins and opened into two distinct methods of discussing what transhumanism could become. Arthur C. Clarke's *2001: A Space Odyssey* crystallized the first: a transhumanism of cosmic transcendence, shedding the social planning of Wells and the biological proposals of Haldane in favor of something mystical and intangible—a story of humanity dissolving into something beyond itself, told not through scientific treatises but through cinema and mass culture. FM 2030 represents the second: a transhumanism of humanitarian rationalism, global in scope, anti-nationalist in impulse, and committed to the idea that technology could liberate the individual from the constraints of culture, poverty, and biological limitation. Where Clarke looked outward to the stars, FM 2030 looked inward at humanity's earthly condition, imagining the same technological potential through the lens of a social planner, an intellectual tradition stretching back to Wells and the Huxley's, now applied without borders.

Ultimately Clarke's cosmic vision became intertwined with transhumanism, while 2030's humanitarian future faded into relative obscurity, stripped of his substance and originality as he was absorbed into the Extropian canon. In doing so, this chapter also marks a threshold in transhumanist discourse itself: the moment the movement split between a transhumanism of cosmic transcendence—concerned with what humanity might become beyond biology, beyond earth, beyond comprehension—and a transhumanism of social liberation, concerned with the earthly conditions that prevented humanity from flourishing in the present. Both strands feed into contemporary transhumanism, but they have not been treated equally: the cosmic vision dominates, while the humanitarian one has been largely forgotten.

Scholars studying transhumanism, ranging from philosophers to social scientists, have increasingly identified modern transhumanism, particularly in its Silicon Valley iterations, as functioning less as a scientific philosophy than as a secular religion. Hava Tirosh-Samuelson has argued that transhumanism operates as a "secularist faith," offering technological substitutes for theology's core promises: immortality through cryonics and mind uploading, omniscience through artificial superintelligence, heaven through virtual reality.¹³⁹ Meghan O'Gieblyn's widely cited essay, *Ghost in the Cloud*, has traced the theological genealogy even more precisely connecting Silicon Valley's transhumanist faith directly to Christian eschatology—the Singularity as a technological Second Coming, digital resurrection as a scientific Last Judgment, the merge with superintelligent AI as a form of apotheosis.¹⁴⁰ Roy Jackson's *Muslim or Supermuslim* explains how the transhumanist scientist creed has been deified.¹⁴¹ Academic scholarship has further identified modern transhumanism as converging with the science-oriented wing of the New Age movement, providing what one peer-reviewed study calls "a new eschatological scenario" for an audience that had abandoned traditional religion without abandoning the desire for transcendence. I agree with these religious assertions and argue that their frameworks hold in the history of transhumanism, finding themselves fully formed within Clarke's *2001*, partially inspired by Stapledon, and going back as far as H.G. Wells.

The world Clarke and F.M. Esfandiary wrote in was defined by rapid change and a set of historical pressures unique to a post-World War II world. The nuclear warheads possessed by both the United States and the USSR, enough to threaten total destruction, transformed a peak of

¹³⁹ Tirosh-Samuelson, "Transhumanism as a Secularist Faith."

¹⁴⁰ Meghan O'Gieblyn, "Ghost in the Cloud," *n+1*, April 2017, <https://www.nplusonemag.com/issue-28/essays/ghost-in-the-cloud/>.

¹⁴¹ Jackson, *Muslim or Supermuslim* 163-174.

scientific triumph into an existential threat. The 1962 Cuban Missile Crisis crystallized this tension, bringing the world to the edge of annihilation. Experts have long contested the psychological effects these existential threats had on people, but most experts agree that generally, people felt apathetic as nuclear war was something so large and abstract that they themselves were not capable of influencing the outcome.¹⁴² Similarly, the era of European world dominance was over, the continent instead divided between USSR and American hegemony. Imperial structures were collapsing across Asia and Africa, further weakening imperialist European powers.

Through Clarke and Esfandiary, the intellectual seat of transhumanism migrated from England to a global stage. While Clarke was an English man, he lived in Sri Lanka by the sixties and for the rest of his life, and Esfandiary's nomadic life found permanent residence in California. Sixties civil rights movements were demanding that the democratic promises of Western nations be extended to all their citizens, and Second-wave feminism was challenging gendered hierarchies throughout the world. Science stood caught between its greatest triumphs—the space race, the birth control pill, the green revolution—and its most visible catastrophes: DDT, nuclear testing, and the slow-dawning recognition that technological progress and human flourishing were not intrinsically the same. Rick Perlstein, an esteemed author of American history, says that in the 1968-1979 period “America suffered more wounds to its ideal of itself than at just about any other time in its history.”¹⁴³ This was the historical atmosphere in which

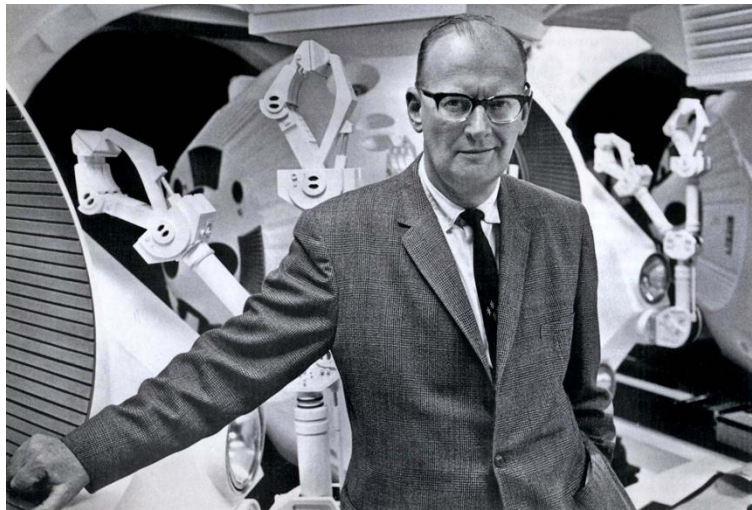
¹⁴² Institute of Medicine (US), Steering Committee for the Symposium on the Medical Implications of Nuclear War, "Adult Beliefs, Feelings, and Actions Regarding Nuclear War: Evidence from Surveys and Experiments," in *The Medical Implications of Nuclear War* (National Academies Press, 1986), <https://www.ncbi.nlm.nih.gov/books/NBK219181/>.

¹⁴³ Rick Perlstein, *The Invisible Bridge: The Fall of Nixon and the Rise of Reagan* (Simon & Schuster, 2014).

Clarke asked what humanity might transcend into, and FM Esfandiary asked what humanity needed to be freed from.

Section II: Arthur Charles Clarke.

Arthur Charles Clarke was born in Minehead, Somerset in 1917, and discovered science fiction at age thirteen through the pages of the *Astounding Stories* magazine—an encounter he described in his autobiography as the beginning of a lifelong devotion.¹⁴⁴ He served in the Royal Air Force during World War II as a radar instructor; he would write *The Rocket and the Future of Warfare* during his time, writing in his biography that he echoed H.G. Wells' antiwar sentiments.¹⁴⁵ Clarke was an accomplished technological futurist as well as a science fiction author; he is widely credited with first proposing the concept of geostationary communications



satellites in a 1945 technical paper, a technology that has since fundamentally changed the world as the base of the internet.¹⁴⁶

Figure 8. Arthur C. Clarke during the filming of *2001: A Space Odyssey* (1968). 1968 Metro-Goldwyn-Mayer Inc. Sourced from <https://www.britannica.com/biography/Arthur-C-Clarke#/media/1/120003/212495>

¹⁴⁴ Arthur C. Clarke, *Astounding Days: A Science Fictional Autobiography* (Gollancz, 1989), 10–13.

¹⁴⁵ Clarke, *Astounding Days*, 39, 98, 181.

¹⁴⁶ Arthur C. Clarke, "Extra-Terrestrial Relays: Can Rocket Stations Give World-Wide Radio Coverage?," *Wireless World*, 1945, <http://clarkeinstitute.org/wp-content/uploads/2010/04/ClarkeWirelessWorldArticle.pdf>.

Clarke positioned himself as the intellectual heir of the proto-transhumanist lineage this thesis traces. In his autobiography *Astounding Days*, he recalls "my first encounter with the master, H.G. Wells," where he used his lunch hour every day to discretely read a copy of Well's *The War of the Worlds* at a local library.¹⁴⁷ Clarke cited Wells as his first serious science fiction author, and notes that he appeared to have "followed in Wells's footsteps—quite unconsciously—not for the first time."¹⁴⁸ Olaf Stapledon received even greater reverence: Clarke wrote "no book before or since *Last and First Men* had such an impact on my imagination," and twenty years after first reading it, he persuaded an elderly Stapledon to address a gathering on the social and biological aspects of space exploration.¹⁴⁹ UNESCO formalized an institutional connection in 1961 by awarding Clarke the Kalinga Prize—an award previously given to both Bertrand Russell and Julian Huxley, linking Clarke directly to the proto-transhumanist lineage through the same institution Julian Huxley had helped found.¹⁵⁰ Clarke wrote over fifty novels and, to paraphrase a sixties newspaper, "four jillion" short stories and articles.¹⁵¹ He was, by any measure, the most widely read figure in this entire study: one reviewer called him "without question the world's best-known and bestselling science fiction writer."¹⁵² *2001: A Space Odyssey*, co-written with Stanley Kubrick, was the second highest-grossing film of 1968 and was crowned the greatest film ever made in a 2022 poll of 480 directors.¹⁵³

¹⁴⁷ Clarke, *Astounding Days*, 28.

¹⁴⁸ Clarke, *Astounding Days*, 181.

¹⁴⁹ Clarke, *Astounding Days*, 29. He also referred to Stapledon as "one of the noblest and most civilized minds," and expressed joy over Stapledon's revival in the late 80s.

¹⁵⁰ "Laureates," UNESCO, <https://www.unesco.org/en/prizes/popularization-science/laureates>.

¹⁵¹ Victor Cohn, "Idea for New Space Movie Came From a Fertile Mind," *Washington Post*, March 31, 1968.

¹⁵² Clarke, *Astounding Days*, 232.

¹⁵³ Bob McClay, "2001: A Space Odyssey," *Rolling Stone*, July 6, 1968; Calum Russell, "Stanley Kubrick's '2001' Named Greatest Movie of All Time," *Far Out Magazine*, December 2, 2022, <https://faroutmagazine.co.uk/2001-a-space-odyssey-named-greatest-movie-of-all-time/>.

The scholarly literature on Clarke is vast but lopsided. As one of the so-called "Big Three" science fiction writers alongside Isaac Asimov and Robert Heinlein, Clarke has received extensive treatment in science fiction scholarship, and *2001* in particular has been examined as a cinematic and literary achievement of the first order. What the scholarship has not done is treat Clarke seriously as a transhumanist thinker in the intellectual lineage traced by this thesis. The transhumanist histories that mention him—Bostrom, Bohan—do so briefly, acknowledging his popularizing role without examining how his work shaped the movement's core assumptions about transcendence, masculinity, and the relationship between humanity and technology. Furthermore, the few transhumanist histories that do examine Clarke, such as Robert Geraci's "There and Back Again" do not link him to prior transhumanist authors.¹⁵⁴ Given Clarke's significance as perhaps the most influential science fiction author of all time, I argue the stakes of him fitting into transhumanist canon are large, as his influence reached into every corner of science fiction, and by extension, transhumanism.

The contemporary reception of Clarke is worth examining. Clarke, interviewed by Victor Cohn in 1968 before *2001* was released, described the film's central idea: "We're part of an evolution from apes to angels. And the question is, where are we at the moment?"¹⁵⁵ The film's initial reception was divided—some critics hailed it as a visionary masterpiece while others found it ponderous and opaque—but it found its most passionate audience among the countercultural youth, who returned to it repeatedly, often under the influence of psychedelics,

¹⁵⁴ Robert M. Geraci, "There and Back Again: Transhumanist Evangelism in Science Fiction and Popular Science," *Implicit Religion* 14, no. 2 (2011): 141–47, <https://doi.org/10.1558/imre.v14i2.141>. Notably, Geraci is

¹⁵⁴... referring to both the popularization of transhumanism and Clarke's role in infusing transhumanism with a religious element.

¹⁵⁵ Cohn, "Idea for New Space Movie Came From a Fertile Mind."

drawn to its mystical themes.¹⁵⁶ Clarke's work received reviews from just about anyone or any news outlet writing movie reviews. Within this chapter, Clarke receives greater attention than F.M. Esfandiary, as Clarke's impact within science fiction and transhumanism was titanic. Furthermore, I briefly examine *2010* throughout my analysis. *2010* was not written in collaboration with Stanley Kubrick, and is a continuation of *2001*, allowing us to see Clarke independently continue the patterns I identify in *2001*.

Section III: Clarke and Gender

Clarke continued with the gendered hierarchies scattered throughout proto-transhumanist tradition and extended them further than any of his predecessors. He did this not as explicit ideology, as in Wells, nor as accidental implication, as in Haldane but as a structural principle woven into the fabric of the future he imagined. *2001*'s prehistoric opening establishes the pattern when a mysterious Obelisk descends, millions of years before civilization, to select man-apes for enlightenment, which entails teaching them to throw stones, fashion tools, and hunt. Despite the near universality of this instruction, one group is conspicuously absent: females. Clarke refers to every man-ape selected for transformation with masculine pronouns throughout; female man-apes appear only as mothers or injured figures requiring assistance, possessing no capacity for independent action.¹⁵⁷ Clarke offers no commentary on this exclusion. It is not presented as a choice but as a fact of the universe—transcendence, from the very origin of human potential, is masculine. The pattern holds three million years later, when the novel's

¹⁵⁶ Joseph Gelmis, "'Space Odyssey' Fails Most Gloriously," *Newsday*, April 4, 1968; Moira Walsh, "2001: A Space Odyssey," *America*, April 27, 1968; McClay, "2001"; Jim Watters, "'2001: A Space Odyssey,'" *Boxoffice*, April 8, 1968; John Coleman, "2001: A Space Odyssey," *New Statesman*, 1968; "2001: A Space Odyssey," *Monthly Film Bulletin*, 1968; Seth Friedman, "Beyond the Infinite Interpretations: The Reception of 2001: A Space Odyssey and the Reframing of Stanley Kubrick's Authorial Reputation," *Film History* 32, no. 1 (2020): 72–105.

¹⁵⁷ Arthur C. Clarke, *2001: A Space Odyssey* (New American Library, 1968), 8–32.

human characters reach for the stars. Female scientists on the moon are fundamentally "girl assistants," despite no apparent difference in credentials or capability.¹⁵⁸

In 1963, Betty Friedan, an influential feminist, wrote *The Feminine Mystique*; by the end of the year, second wave feminism in America was alive. *The Feminine Mystique* identified a problem plaguing American women, particularly white housewives, known as “the problem with no name.”¹⁵⁹ The problem with no name refers to a collective societal ignorance of women’s needs for fulfillment beyond that from being a mother and wife: Friedan writes “where is the world of thoughts and ideas, the life of the mind and spirit?”¹⁶⁰ In *2001*, women appear purely in service roles, stewardess and domestic partners, confined to what Clarke's universe treats as their natural sphere: earth, embodiment, and service to men. Even the superintelligent AI, HAL, is coded as male. By 1968 Second-wave feminists had a series of legal victories, had been recognized by two presidents, had organized into the National Organization for Women, and after a three-year battle, won their inclusion in the civil rights act of 1964.¹⁶¹ Second-wave feminism was not quiet; it was roaring. It is implausible Clarke would not have been presented with the feminist arguments that were reaching every corner of society. In *2001*, by the end of the first act, the only feminine entities left are smaller space pods aboard the ship, “usually christened with feminine names, perhaps in recognition of the fact that their personalities were...unpredictable.”¹⁶² Clarke’s gendered hierarchy was not an inevitable product of his time. He extends this hierarchy backward to the prehistoric origin of intelligence and forward to

¹⁵⁸ Clarke, *2001*, 62.

¹⁵⁹ Betty Friedan, *The Feminine Mystique* (1963; W. W. Norton, 1963), 1-29.

¹⁶⁰ Friedan, *Feminine Mystique*, 32.

¹⁶¹ Allen Fisher and Lyndon B. Johnson Presidential Library and Museum, “Women’s Rights and the Civil Rights Act of 1964,” National Archives, July 3, 2019, <https://www.archives.gov/women/1964-civil-rights-act>.

¹⁶² Clarke, *2001*, 119.

artificial superintelligence suggests something more than just the unreflective sexism of his time. Femininity, in Clarke's world, is not merely a social condition with social consequences, but instead a cosmic position.

Beyond simply excluding women from transcendence, Clarke reproduced the Wellsian framework in which femininity actively threatens male rational achievement. Clarke revealed the most direct line of inheritance between his work and the proto-transhumanist tradition this thesis traces. In *2010*, a male crew member offers an unprompted illustration of the danger: one of the brightest engineers in his class, he explains, was "fatally attracted to a blonde in Kiev" and ended up working in a garage despite winning a gold medal for designing space stations.¹⁶³ The parallel to Wells is unmistakable—a clever woman diverts a gifted man from his appointed rational work, the same fear that animated Wells's warnings about women and capitalism in *Anticipations*. Even the term used for the disrupted man—engineer—is the same as in *Anticipations*. Through Clarke's biography we know that he read Wells and consumed not just the fiction of the early 1900s, but we also know he read non-fiction social planning works like J.D. Bernal's *The World*.¹⁶⁴

The most analytically significant version of Clarke's gendered hierarchy involves not a human woman but an artificial intelligence: HAL 9000. HAL, the male AI of *2001*, is paranoid, murderous, and catastrophically unstable—and yet it is HAL who achieves transcendence in *2010*, evolving into an energy-based lifeform of cosmic significance.¹⁶⁵ SAL, HAL's female counterpart, exhibits precisely the stability and rationality HAL lacks, and is entirely excluded

¹⁶³ Arthur C. Clarke, *2010: Odyssey Two* (Del Rey Books, 1982), 78.

¹⁶⁴ Arthur C. Clarke, *Greetings, Carbon-Based Bipeds!: Collected Essays, 1934–1998* (St. Martin's Griffin, 2000),

¹⁶⁵ Arthur C. Clarke, *2010*. 79

from any mission or opportunity for discovery or transcendence. Clarke's universe places gender above rationality as the determinant of destiny. This is a significant departure from Wells, Haldane, and the Huxleys, who justified consciously or subconsciously, male dominance through rational superiority. Clarke does not bother with that justification. In his cosmos, a volatile, murderous male AI earns transcendence that a calm, competent female AI does not. Femininity is non-transcendent not because it lacks rationality but because Clarke's universe has decided that it simply is. Other than Stapledon, Clarke is the only author examined in this thesis to put something above both rationality and science in importance—gender.

Section IV: Clarke and Spirituality

Clarke's *Odyssey* novels are full of religious imagery; it makes up the very backbone of the *Odyssey* novels. The Obelisk functions as a god—omniscient, inscrutable, selecting the worthy for enlightenment and guiding human evolution across millions of years with a purpose it never deigns to explain. HAL and Dave Bowman's dissolution into energy-based lifeforms parallels the theological concept of the soul shedding mortal flesh, transcendence not through biological improvement, as Haldane imagined, nor through mechanical augmentation, as Bernal proposed, but through the abandonment of the physical entirely.¹⁶⁶ The creation of a second sun in *2001* is an act of literally biblical creative power, invoking the Genesis narrative of the bible. As this thesis has demonstrated, a religious framework for transhumanism and futurism is rich in intellectual history, tying back to Wells' sacred lantern, Stapledon's cosmic spiritualism, and Julian Huxley's evolutionary humanism, each of which preserved the emotional and moral architecture of religion while displacing its theological foundations.

¹⁶⁶ Clarke, *2001*, 214–21.

Clarke's debt to Stapledon is direct and acknowledged. Clarke wrote that *Last and First Men* had an unparalleled "impact on my imagination," and the fingerprints of Stapledon's *Last and First Men* are visible throughout the *Odyssey* novels.¹⁶⁷ For instance, in *Last and First Men*, advanced alien intelligence that guides human evolution across millions of years, the capacity to enter other minds, the drive to seed life across a dying solar system, the dissolution of individual consciousness into something collective and cosmic.¹⁶⁸ Clarke's aliens are Stapledonian in nature: an enlightened hive mind, simultaneously individual and collective, pursuing the spread of consciousness across the universe as an end worthy in itself.¹⁶⁹ Where Stapledon's *Last Men* reached backward through time to cultivate Jesus, Socrates, and Gautama, Clarke's aliens influenced the ancient past by selecting and elevating the worthy among primitive man-apes.¹⁷⁰ Both visions imagine transcendence as a guided process, overseen by beings of incomprehensible advancement, moving toward a destination that rational language cannot describe.

The 1960s represented a widespread cultural turning away from the rationalist certainties that had defined the first half of the twentieth century. The thinkers of chapters one and two had written in a world defined by depression, war, and the conviction that science and rational planning were humanity's most reliable tools. Clarke wrote in a world that had watched DDT poison ecosystems, nuclear testing irradiate the atmosphere, and the most technologically advanced military in history fail to translate firepower into meaning in Vietnam. The counterculture responded to this disillusionment not with more rationalism but with its opposite:

¹⁶⁷ Clarke, *Astounding Days*; Clarke, 2010; Clarke, 2001.

¹⁶⁸ Stapledon, *Story of the Near and Far Future*, 270-294

¹⁶⁹ Clarke, 2010, chapters 46-47; Clarke, 2001, chapters 37-38.

¹⁷⁰ Clarke, 2001: *A Space Odyssey*. 8-32

the hippie movement, an explosion of interest in Eastern spirituality, astrology, and the paranormal all represented a mass cultural reaching for frameworks of meaning that science and institutional planning had demonstrably failed to supply. The Hartford Institute for Religion Research documents a considerable growth in astrology and mystical belief as a direct consequence of 1960s counterculture.¹⁷¹ Furthermore, the film version of *2001* explicitly took advantage of this, marketing itself as “the ultimate trip,” tapping further into the counterculture.¹⁷² Clarke’s fiction arrived when this spiritual hunger was at its peak, and it spoke directly to an audience that craved transcendence beyond what rationalism could offer.

When we look at both the role of gender and spirituality in *2001*, we find two statements at the heart of Clarke’s work: Transcendence is the future, and women are not meant for transcendence. These ideas were historically situated in both Clarke’s past and present, and for many people, was their first invitation to think about the future—not the future of a nuclear war, or what job or who they would marry, or even the future of politics and governance on earth, but the future of *existence*. Clarke’s film is so successful that it still demands cultural respect, showing up in *Star Wars*, as an easter egg on apple iPhones, in the most popular cartoons, and even adverts for the immensely successful *Barbie* (2023) movie.¹⁷³ The ideas formulated by Wells, Haldane, Bernal, Stapledon and the Huxleys were revitalized and tailored for a new audience and time through Clarke. The commanding presence of *2001* in the modern day coupled with its connections to the past means these thinkers theories and hierarchies are not just

¹⁷¹ Colin Campbell, “Astrology,” in *Encyclopedia of Religion and Society*, ed. William H. Swatos Jr. (AltaMira Press, 1998), <https://hrr.hartfordinternational.edu/articles/4141/>.

¹⁷² Friedman, *Beyond the Infinite Interpretations*, 71–92.

¹⁷³ Nathan Abrams, “2001: A Space Odyssey Still Leaves an Indelible Mark on Our Culture 55 Years On,” *The Conversation*, July 10, 2023, <https://doi.org/10.64628/AB.7vkmygddh>.

charming or horrifying relics of the past, but living ideas, the basis that transhumanists of today build off of.

Section V: F.M. 2030: A Humanistic Transhumanism.

FM 2030 was born Fereidoun M. Esfandiary in BRussells in 1930, the son of an Iranian diplomat. His childhood was nomadic by design—he attended primary school in both Iran and London before moving to a Jesuit school in Jerusalem; he had lived in 17 countries by age 11.¹⁷⁴ This was a trajectory that gave him fluency in multiple languages and a profound estrangement from any single national identity that would define his thought for the rest of his life.¹⁷⁵ He competed in the 1948 Olympic Games representing Iran, earned a degree from the University of California in 1952, and worked as a United Nations diplomat for two years—a role that, like Julian Huxley's UNESCO directorship, gave him both a global career and an institutional grounding for his futurist ambitions.¹⁷⁶ He later became a professor, writer, and prominent figure in the West Coast futurist networks that would eventually produce the Extropian movement.¹⁷⁷ He renamed himself to F.M. 2030 in the seventies because of his conviction that 2030 would be a “magical time,” where aging had been cured and people were free of every restriction.¹⁷⁸ I will refer to him as 2030 henceforth. Upon his death in 2000, FM 2030 was cryonically frozen, in the hope of waking in the year 2030.¹⁷⁹ This renaming was a philosophical statement: he rejected the

¹⁷⁴ “F. M. Esfandiary / FM-2030 Papers,” Manuscripts and Archives Division, New York Public Library.

¹⁷⁵ “F. M. Esfandiary / FM-2030 Papers,” Manuscripts and Archives Division, New York Public Library.

¹⁷⁶ “F. M. Esfandiary / FM-2030 Papers.”

¹⁷⁷ Benjamin Tiven, “The Future Takes Forever: Becoming FM-2030,” Bidoun, n.d., <https://bidoun.org/articles/the-future-takes-forever>; James J. Hughes, “The Politics of Transhumanism and the Techno-Millennial Imagination, 1626–2030,” *Zygon* 47, no. 4 (2012): 757–76, <https://doi.org/10.1111/j.1467-9744.2012.01289.x>.

¹⁷⁸ F. M. 2030, *Are You a Transhuman? Monitoring and Stimulating Your Personal Rate of Growth in a Rapidly Changing World* (Warner Books, 1989), 2.

¹⁷⁹ Fred Chamberlain, “Cryonics: FM-2030 Now Hurtling into the Future,” *Cryonics* (Alcor Life Extension Foundation), 2000, <https://web.archive.org/web/20101119134918/http://alcor.org/cryonics/cryonics2000-4.pdf>.

nationality and biological identities that a traditional name brought, instead embodying in his own person the transhumanist arguments he made in his books.



Figure 9. FM 2030 (Fereidoun M. Esfandiary). Photograph.
Wikimedia Commons.
<https://commons.wikimedia.org/wiki/File:FM2030.jpg>.

FM 2030 has received almost no scholarly attention. He appears as a named figure in James Hughes's *The Politics of Transhumanism* and in scattered Extropian histories, but his novels remain essentially unexamined academically. In the seventies, his reception varied from book to book. *Identity Card* (1969) was received with overwhelmingly positive reviews, a high volume of reviews, and a marker of rarity: a novel about the Middle East written not by a

Western observer but from the inside, by an Iranian whose perspective on his own culture carried an authenticity that Western expatriate fiction could not replicate.¹⁸⁰ The reception of *Optimism One* (1970) was mixed. Some reviewers found *Optimism One* topical and genuinely refreshing, a breath of optimism in a moment when public discourse was dominated by pessimism and apathy.¹⁸¹ One author found it tedious, lacking in citations and requiring the reader to blindly trust Esfandiary.¹⁸² Regardless, the novel arrived with little fanfare, at least that which took the form of written reviews. This thesis's analysis of his writings and my conception of FM 2030 as an early transhumanist outside of the mainline canon is wholly original. Viewed through the outside-main-canon lens of my thesis, 2030 serves as a significant representation of non-Extropian transhumanism beginning prior to Extropians and stretching well into their most dominant era.

FM 2030 carried on the transhumanist tradition of a social planner's framework, as well as the conviction that rationalism was the supreme organizing principle of an optimal society, the belief that science and technology were humanity's most reliable tools for human liberation, and that the future belonged to those willing to shed the emotional, traditional, and sentimental attachments of the past. What he transformed was the geography of that framework. This thesis asserts that where Wells had mapped the rational-irrational divide onto class lines within Britain, FM 2030 mapped it onto entire nations and cultures. In *Identity Card*, his protagonist Daryoush—a transparent self-insert—declares that the age of romanticism is over, that there is

¹⁸⁰A collection of reviews to back this up: "Book Notes." *Hollywood Reporter*, 1965, 188, no. 28, 6. Esfandiary, F. M. "Fun-house Hell." *Newsweek*, October 10, 1966, 122. Esfandiary, F. M. "The Mystical West Puzzles the Practical East." *New York Times*, February 5, 1967. F.L. "Buck-Passing Unlimited." *New York Times*, October 2, 1966. Kriegel, Leonard. "Private & Public Sensibility." *Nation* 203, no. 19 (1966): 616–20. T. L. "Books of the Times: Strangled." *New York Times*, November 8, 1966.

¹⁸¹ Ozmon, Howard. Review of *Optimism One: The Emerging Radicalism*, by F. M. Esfandiary. *Phi Delta Kappan* 52, no. 9 (1971): 562. <http://www.jstor.org/stable/20373016>.

¹⁸² Claiborne, Robert. "Stoned on Technology." *Washington Post, Times Herald*, March 7, 1971, 4.

no room for poetry in an age of science, and that only those willing to embrace rational realism have any claim on the future.¹⁸³ This section argues *Identity Card* codes Iran's rich culture of poetry and art as passive, emotional, and archaic.¹⁸⁴ The mechanism of this cultural hierarchy is language: the modern man speaks English and French, while the man of the "Middle Ages" uses Arabic and Turkish.¹⁸⁵ Modernity is Western, and modernity is rational. I interpret *Optimism One* as an extension of this framework, coding Asia, Africa, and Latin America in similarly passive terms: entire continents positioned as obstacles to rational modernity rather than participants in it. 2030 uses harsh diction for these societies to demarcate them as primitive, such as their "pissing in pots" to their "mud huts," "backwater Asian men and African women," and their use of "execution squares."¹⁸⁶

2030 constructed this cultural hierarchy at a moment when global movements challenged such structures. By the 1960s, decolonization had made massive strides and was proceeding at a rapid rate. Franz Fanon's *The Wretched of the Earth* (1961), published five years before *Identity Card*, argued with that colonialism was not merely a political arrangement but a psychological one, which caused the colonized had to feel inferior, archaic, and passive.¹⁸⁷ This chapter contends this psychological process is exactly what F.M. 2030 was advocating for. Kwame Nkrumah, a Ghanaian political theorist and revolutionary wrote *Neo-Colonialism: The Last Stage of Imperialism* (1965), which extended this psychological argument into the economic realm and warned that formal independence without economic sovereignty merely replaced old imperialism

¹⁸³ 2030, *Identity Card*, 35, 41, 50–70, 97, 100–101, 123, 205.

¹⁸⁴ 2030, *Identity Card*, 50–70.

¹⁸⁵ 2030, *Identity Card*, 89–90.

¹⁸⁶ F. M. 2030, *Optimism One: The Emerging Radicalism* (Norton, 1970), 35, 41, 97, 100–101, 123, 205.

¹⁸⁷ Frantz Fanon and Constance Farrington, *The Wretched of the Earth* (Grove Press, 1961).

with new.¹⁸⁸ FM 2030 must have been aware of these arguments; he had worked as a UN diplomat, moved in international circles, and positioned himself explicitly as an anti-nationalist and a global citizen. Despite this, a close reading of his framework reproduces the very civilizational hierarchy that decolonization movements were dismantling: non-Western cultures coded as passive, emotional, and archaic, Western rationalism as the universal standard, tourism as the mechanism of uplift.

FM 2030's cultural hierarchy operated alongside a familiar gendered one, and the two were inseparable. In *Identity Card*, women are divided into two categories: "simple" women who offer physical and emotional satisfaction, and "complex" women who provide intellectual companionship; this divide maps directly onto 2030's East/West hierarchy.¹⁸⁹ Iranian women embody passive, archaic femininity, while Western women approach masculine rationality through cosmopolitan mobility.¹⁹⁰ Safoura, the Iranian woman with whom Daryoush has a brief sexual relationship, is praised for her body while her mind is dismissed entirely—she accepts domestic violence as inevitable, remains "resigned to her medieval fears and sacred superstitions," and has "subdued within herself all ambition, all drive, and aggressiveness."¹⁹¹ The ideal woman, by contrast, has traveled widely and nourishes "a broad, rational view of life"—yet even she exists in relation to male intellectual work, needing only to "perceive and enjoy" male creativity rather than produce her own.¹⁹²

¹⁸⁸ Kwame Nkrumah, *Neo-Colonialism: The Last Stage of Imperialism* (Panaf, 2004).

¹⁸⁹ 2030, *Identity Card*, 120–27.

¹⁹⁰ 2030, *Identity Card*, 120–27.

¹⁹¹ 2030, *Identity Card*, 118–35.

¹⁹² 2030, *Identity Card*, 120–25.

The gendered hierarchy of early transhumanism is structurally intact: women, elevated ones included, serve as audiences and companions to rational men rather than intellectual agents in their own right. This was the earliest and most hierarchical point in FM 2030's career, and his thought evolved considerably. By *Up-Wingers*, the gendered framework had transformed. He wrote: "How free am I if I cannot choose my own body my own brain my own gender?"—framing gender itself as a constraint to be overcome rather than a category to be ranked.¹⁹³ He pushed further still, asking "Why know only this specific gender? Why not the other gender or an alternation between both genders or a fusion of the two?"¹⁹⁴ He imagined a future in which race, gender, height, and skin color become "irrelevant" as biological modification renders fixed identity obsolete.¹⁹⁵

FM 2030's evolution on gender anticipates a tension that would come to define modern transhumanism. His move from the explicit gendered hierarchy of *Identity Card* to the post-gender universalism of *Up-Wingers* mirrors the broader transhumanist trajectory: rather than reckoning with existing gender inequality, the movement dissolved the category entirely, declaring gender an obsolete biological constraint to be transcended through technology. Dvorsky and Hughes's foundational 2008 transhumanist essay "Postgenderism: Beyond the Gender Binary" explicitly frames postgenderism as rejecting "ameliorating inequalities within existing gender roles in favor of their technological obsolescence."¹⁹⁶ The hierarchy is not dismantled; it is declared irrelevant, and the question of who occupied which position within it before the dissolution is never asked. I argue that FM 2030 was among the earliest transhumanist

¹⁹³ F. M. 2030, *Up-Wingers* (John Day, 1973), 55.

¹⁹⁴ 2030, *Up-Wingers*, 62.

¹⁹⁵ 2030, *Up-Wingers*, 63.

¹⁹⁶ George Dvorsky and James Hughes, *Postgenderism: Beyond the Gender Binary* (Institute for Ethics and Emerging Technologies, 2008), <https://doi.org/10.13140/RG.2.2.18028.08324>.

thinkers to make this move—asking "How free am I if I cannot choose my own body my own brain my own gender?"¹⁹⁷ History carries baggage, and failing to face gender inequalities sweeps the implicit hierarchies designated into the design of the future under the rug. Modern transhumanism remains, "only of interest to cisgender white men" despite its post gender aspirations.¹⁹⁸

Section VI: Concluding Thoughts.

What the Extropian movement inherited from Clarke was cosmic myth: A vision of humanity dissolving into something beyond itself, guided by forces of incomprehensible intelligence, unconstrained by politics or social obligation. What it inherited from FM 2030 was his framework without the humanity: The UpWinger imperative to move forward and upward, stripped of the social planning, the anti-nationalism, and the collective liberation that FM 2030 had insisted were inseparable from the transhumanist project. The humanitarian transhumanism FM 2030 represented has never been recognized as the independent and original formation of thought that it is. That selective appropriation is not incidental but structural, a pattern this thesis has traced from Bohan's omission of Stapledon to Bostrom's removal of Russell and dismissal of FM 2030 as a "nose job" celebrity rather than a transhumanist on early Extropian forums.¹⁹⁹ Transhumanist history has consistently chosen the cosmic over the earthly, the individual over the collective, the transcendent over the political—and in doing so has shaped not just how the movement remembers its past, but what it believes its future can contain.

¹⁹⁷ 2030, Up-Wingers, 55.

¹⁹⁸ Charlotte Shane, "Life Extension Technology Gives Us a Bleak Future: More White Men," Jezebel, 2016, <https://www.jezebel.com/life-extension-technology-gives-us-a-bleak-future-more-1793857274>.

¹⁹⁹ Bostrom, "History of Transhumanist Thought"

The questions these thinkers argued about—what humanity might become, who gets to lead that transformation, by what values and for whose benefit—have not been settled. They have only been obscured, buried beneath the weight of a movement that has convinced itself its answers are inevitable. Recovering the alternatives, the roads not taken, the voices dismissed and the arguments erased, is not merely an academic exercise. It is a reminder that the transhumanism that exists today is not the only transhumanism that was possible, and that the choices made in drawing rooms, laboratories, UNESCO committees, science fiction magazines, and Extropian mailing lists were choices, not destinies. Understanding them as such is where any honest history of the movement must begin.

Concluding Thoughts: Looking Into the Future of Transhumanism

The 1980s and 1990s witnessed the birth of the first explicitly transhumanist community: the Extropians. Founded in the late 80s by Max More and Tom Morrow, the group championed a volatile blend of Nietzschean philosophy, radical libertarianism, and technological optimism, and most critically, the pursuit of transhumanism.²⁰⁰ What began as an online forum eventually evolved into a monthly magazine, establishing the Extropians as the movement's first theory-oriented transhumanist community.²⁰¹ Many of the most famous names within transhumanism—including Marvin Minsky, Nick Bostrom, Max More, Eliezer Yudkowsky, and Natasha Vita-More—engaged with this early mailing list.²⁰²

By 2006, transhumanist circles had abandoned informal mailing lists and local parties in favor of organization. The Extropy Institute's board of directors dissolved the organization that year, claiming its goals were met as transhumanism reached a broader audience and new groups emerged.²⁰³ The World Transhumanist Organization was created in 1998 by Nick Bostrom and David Pearce; they were both prominent Extropians at the time. The WTA, rebranded to Humanity+ in 2008, became the dominant force in the movement to this day.²⁰⁴ This transition marked the beginning of a libertarian and far-right effort to expel the left, though it did not remove the movement's existing Extropian roots. Traces of that original influence remain;

²⁰⁰ Timnit Gebru and Émile P. Torres, "The TESCREAL Bundle: Eugenics and the Promise of Utopia through Artificial General Intelligence," *First Monday*, April 14, 2024, <https://doi.org/10.5210/fm.v29i4.13636>.

²⁰¹ Hughes, "Politics of Transhumanism," 5–9.

²⁰² "Extropy Institute Directors & Advisors," Extropy Institute, accessed November 17 2025, <https://www.extropy.org/emaillists.htm>; Keach Hagey, "How Peter Thiel's Relationship With Eliezer Yudkowsky Launched the AI Revolution," *Wired*, May 20, 2025, <https://www.wired.com/story/book-excerpt-the-optimist-open-ai-sam-altman/>.

²⁰³ Bostrom, "History of Transhumanist Thought"; Hughes, "Politics of Transhumanism," 8–14.

²⁰⁴ Bostrom, "History of Transhumanist Thought"; Hughes, "Politics of Transhumanism," 7.

Humanity+ has utilized an "FAQ" directly derived from Extropian texts for two decades, and both Max and Natasha Vita-More maintain prominent roles on the organization's website.²⁰⁵

In the early 2000s, far-right capital began flooding the movement. Peter Thiel invested millions into transhumanist projects; by 2017, an investigative journalist noted that his influence was “everywhere.... in the space.”²⁰⁶ James Hughes, a veteran of these early organizations, observed that in 2009, libertarians launched a campaign to seize control of the World Transhumanist Association’s Board of Directors and purge the left. The effort succeeded brilliantly, shifting the demographic of Humanity+ from roughly 67% left leaning to a mere 16%.²⁰⁷ Silicon Valley’s broader rightward tilt has accelerated this push, firmly embedding transhumanism within the "accelerationist" right.²⁰⁸

The emergence of far-right elements within transhumanism is not a superficial change; it is a fundamental transformation of the movement’s goals. In the modern era, the "alt-right" has

²⁰⁵ "Membership," Humanity+, accessed March 1, 2026, <https://www.humanityplus.org/membership>;

"Transhumanist FAQ," Humanity+, accessed March 7, 2026, <https://www.humanityplus.org/transhumanist-faq>.

²⁰⁶ Mark O'Connell, *To Be a Machine* (Doubleday, 2017); Alexander Thomas, "A Brief History of Transhumanism and Its Critics," in *The Politics and Ethics of Transhumanism* (Bristol University Press, 2024), <https://doi.org/10.2307/jj.14284473.4>.

²⁰⁷ Hughes, "Politics of Transhumanism," 9–16.

²⁰⁸ Susan B. Levin, "Silicon Valley's Favorite Weird Philosophy Is Fundamentally Wrong," *Slate*, March 22, 2022, <https://slate.com/technology/2022/03/silicon-valley-transhumanism-eugenics-information.html>; Amy Goodman, "Alliance Between Far Right and Silicon Valley Gives Rise to 'End Times Fascism,'" *Truthout*, May 5, 2025, <https://truthout.org/video/alliance-between-far-right-and-silicon-valley-gives-rise-to-end-times-fascism/>; Adam Becker, *More Everything Forever: AI Overlords, Space Empires, and Silicon Valley's Crusade to Control the Fate of Humanity* (Basic Books, 2025); Dan Levin and Craig Nigrelli, "Silicon Valley Leans Historically Left, so Why the Sudden Shift to the Right?," *Straight Arrow News*, August 15, 2025, <https://san.com/cc/silicon-valley-leans-historically-left-so-why-the-sudden-shift-to-the-right/>; Tim Bajarín, "Silicon Valley's Shifting Politics — and The One Thing Uniting Them," *Forbes*, May 20, 2025, <https://www.forbes.com/sites/timbajarin/2025/05/20/silicon-valleys-shifting-politics-and-the-one-thing-uniting-them/>; "Musk and Altman Lead a New Era of Transhumanism with Brain Implants and AI," *BGNES*, August 22, 2025, <https://www.bgn.es.com/science/musk-and-altman-lead-a-new-era-of-transhumanism-with-brain-implants-and-ai>; Nick Robins-Early, "Oxford Shuts Down Institute Run by Elon Musk-Backed Philosopher," *The Guardian*, April 19, 2024, <https://www.theguardian.com/technology/2024/apr/19/oxford-future-of-humanity-institute-closes>; Jonathan Taplin, "How Musk, Thiel, Zuckerberg, and Andreessen — Four Billionaire Techno-Oligarchs — Are Creating an Alternate, Autocratic Reality," *Vanity Fair*, August 22, 2023, <https://www.vanityfair.com/news/2023/08/musk-thiel-zuckerberg-andreessen-alternate-autocratic-reality>.

increasingly adopted right-libertarian frameworks, utilizing cutting-edge technologies, such as social media bot swarms and generative AI, to manufacture consensus and exert political influence. While institutions like Humanity+ often profess a "technoprogressive" or apolitical stance, this facade masks a deep-seated network of far-right capital and influence. The link to Peter Thiel is essential for understanding the movement as not just libertarian, but also alt-right.²⁰⁹ This shift has embedded several regressive ideologies into the movement's core: a neo-colonialist drive for space expansion, a narrow Anglo-Saxon cultural focus, and a dismissive, "fact-over-feeling" posture that uses "rationality" to shield itself from social criticism.²¹⁰ Nick Bostrom was fired from Oxford after it was discovered he engaged in race science in early Extropian chats, which is especially chilling with the movements eugenics filled past and white dominated present. This rightward pivot is particularly striking when contrasted with transhumanism's early socialist history and has a vested interest in covering up the diverse history of transhumanism. This shift has embedded several regressive ideologies into the movement's core: a neo-colonialist drive for space expansion, a narrow Anglo-Saxon cultural focus, and a dismissive, "fact-over-feeling" posture that uses "rationality" to shield itself from social criticism. This rightward pivot is particularly striking when contrasted with transhumanism's early socialist history and has a vested interest in covering up the diverse history of transhumanism.

Thiel is a documented patron of the Dark Enlightenment — the neo reactionary movement founded by Curtis Yarvin and philosopher Nick Land, which openly rejects

²⁰⁹ "Max Chafkin, *The Contrarian: Peter Thiel and Silicon Valley's Pursuit of Power* (Penguin Press, 2021).

²¹⁰ "MIT Extropians: Embrace the Future," accessed via Scribd, March 16, 2026, <https://www.scribd.com/document/250804968/Extropians>. This is an Extropian pamphlet distributed to MIT students pushing against affirmative action, pointing to diversity for diversities sake as slowing down scientific progress. This is a ploy that shows up consistently within Extropian forums, but this is the most infamous example.

democracy, advocates for authoritarian technocratic governance, and envisions a transhuman future reserved for a technological elite. Land's accelerationism imagines the intensification of capitalism and technology until it produces a transhuman master-race; Yarvin's neocameralism imagines governance as a corporation, with citizens as consumers and dissent as a market inefficiency to be eliminated.²¹¹ Thiel has described Yarvin as his "most important connection," and Land's transhumanist accelerationism is pervasive throughout the Silicon Valley networks that now fund and shape the movement.²¹² In the modern era, the "alt-right" has increasingly adopted right-libertarian frameworks, utilizing cutting-edge technologies, such as social media bot swarms and generative AI, to manufacture consensus and exert political influence. While institutions like Humanity+ often profess a "technoprogressive" or apolitical stance, this facade masks a deep-seated network of far-right capital and influence. The ideological stakes of this entanglement became visible as recently as 2026, when Thiel continued to deliver a series of lectures, this time near the Vatican, arguing that opposition to technological progress — AI regulation, environmentalism, arms control — constitutes the work of the Antichrist.²¹³ Thiel owns an AI powered surveillance company, Palantir.

Transhumanism, in my view, is inevitable. Over twenty paralyzed individuals currently use brain-computer interface chips to interact with computers at speeds matching or exceeding

²¹¹ Nick Land, "The Dark Enlightenment," *Urban Future*, 2012, <https://www.thedarkenlightenment.com/the-dark-enlightenment-by-nick-land/>.

²¹² Curtis Yarvin, quoted in Corey Pein, "Mouthbreathing Machiavellis Dream of a Silicon Reich," *The Baffler*, May 19, 2014, <https://thebaffler.com/salvos/mouthbreathing-machiavellis>.

²¹³ Peter Thiel, quoted in Emma Goldberg, "Peter Thiel is Delivering 4 Private Sold-Out Lectures in San Francisco — About the Antichrist," *Fortune*, September 2, 2025, <https://fortune.com/2025/09/02/peter-thiel-antichrist-lectures-commonwealth-club-san-francisco-acts-17-collective-rene-girard/>.

"Peter Thiel Brings His Lectures on the Antichrist to the Vatican's Doorstep," *Associated Press*, March 16, 2026, <https://www.pbs.org/newshour/nation/peter-thiel-brings-his-lectures-on-the-antichrist-to-the-vaticans-doorstep>.

those of the average person.²¹⁴ With the right software, these individuals can use mechanical machines to manipulate the physical world as well.²¹⁵ As technology advances, so too does productivity and intelligence. Artificial intelligence holds the capacity to rewrite how our societies function and advance science and knowledge at a rate unfathomable to ourselves in the present day. To me, the reality of the question is not whether transhumanism arrives, but what it will look like.

The human brain has no true present. It constructs experience from deep memory and sensory input roughly 40 milliseconds old, using both to anticipate what comes next. This is why even neurotypical people occasionally hallucinate. No part of human perception exists in a vacuum separate from the past, and transhumanism is no different. The history it inherits shapes how it is understood and who is allowed within it. A past steeped in linear thinking and the worship of rationalism produces a narrow transhumanism, accommodating rationalists while excluding the spiritual, the emotional, environmentalists, and countless others. The specific history constructed within Extropian and Humanity+ networks narrows it further still: libertarian, predominantly white, intellectual, and male. It does not have to be this way. This thesis has demonstrated the breadth of approaches transhumanism can take. It is not inherently rationalist. It is not inherently white, western, or male. It is not inherently mechanical or expansionist. Transhumanism could be socialist, libertarian, communitarian, spiritual—or something that does not yet have a name. The tradition is not fixed.

²¹⁴ Neuralink, "Two Years of Telepathy," Neuralink Updates, January 29, 2026, <https://neuralink.com/updates/two-years-of-telepathy/>.

²¹⁵ Neuralink, "Two Years of Telepathy," Neuralink Updates, January 29, 2026, <https://neuralink.com/updates/two-years-of-telepathy/>.

Bibliography

Primary Sources

Bernal, J. D. *The World, the Flesh and the Devil*. Kegan Paul, Trench, Trubner, 1929.

Bernal, J. D. to J. B. S. Haldane. January 1944. Haldane Papers. Wellcome Collection.

Bernal, J. D. to J. B. S. Haldane. March 1945. HALDANE/4/20/1/27. Wellcome Collection.
<https://wellcomecollection.org/works/p2sat5p3/items>.

Bradbury, Robert J. "Extropians Forum: Re: Meme Trace: Origin of 'Jupiter-Sized Brains'?" Circa 1999. Extropian Forum Archive. <https://diyhpl.us/~bryan/irc/extropians/www.lucifer.com/exi-lists/extropians.3Q99/2090.html>.

Bradbury, Robert J. "Matrioshka Brains." Unpublished manuscript, July 21, 1999.
<https://gwers.net/doc/ai/scaling/hardware/1999-bradbury-matrioshkabrains.pdf>.

Butcher, Fanny. "Aldous Huxley Pokes Fun at Model T World." *Chicago Daily Tribune*, February 8, 1932.

C. D. M. "Huxley Turns Propagandist." *New York Herald Tribune*, February 7, 1932.

"Circular Letter to All the Peoples of the Earth with Various Drafts of the List of Persons to Whom It Was to Be Sent." University of Liverpool Digital Heritage Lab. Accessed November 23, 2025.
<https://digitalheritagelab.liverpool.ac.uk/Documents/Detail/circular-letter-to-all-the-peoples-of-the-earth-with-various-drafts-of-the-list-of-persons-to-whom-it-was-to-be-sent/81229>.

Clarke, Arthur C. "Extra-Terrestrial Relays: Can Rocket Stations Give World-Wide Radio Coverage?" *Wireless World*, 1945. <http://clarkeinstitute.org/wp-content/uploads/2010/04/ClarkeWirelessWorldArticle.pdf>.

Clarke, Arthur C. *2001: A Space Odyssey*. New American Library, 1968.

Clarke, Arthur C. *2010: Odyssey Two*. Del Rey Books, 1982.

Clarke, Arthur C. *Astounding Days: A Science Fictional Autobiography*. Gollancz, 1989.

Clarke, Arthur C. *Greetings, Carbon-Based Biped!: Collected Essays, 1934–1998*. St. Martin's Griffin, 2000.

Coleman, John. "2001: A Space Odyssey." *New Statesman*, 1968.

Courtney, W. L. "The Novels of H.G. Wells." *New York Times*, 1912.

- Esfandiary, F. M. "Fun-house Hell." *Newsweek*, October 10, 1966, 122.
- Esfandiary, F. M. "The Mystical West Puzzles the Practical East." *New York Times*, February 5, 1967.
- "Extropy Institute Directors & Advisors." Extropy Institute. Accessed November 17, 2025.
<https://www.extropy.org/emaillists.htm>.
- "F. M. Esfandiary / FM-2030 Papers." Manuscripts and Archives Division. New York Public Library.
- F. L. "Buck-Passing Unlimited." *New York Times*, October 2, 1966.
- FM 2030. *Are You a Transhuman? Monitoring and Stimulating Your Personal Rate of Growth in a Rapidly Changing World*. Warner Books, 1989.
- FM 2030. *Identity Card*. Grove Press, 1966.
- FM 2030. *Optimism One: The Emerging Radicalism*. Norton, 1970.
- FM 2030. *Up-Wingers*. John Day, 1973.
- Friedan, Betty. *The Feminine Mystique*. W. W. Norton, 1963.
- Gannett, Lewis. "Books and Things." *New York Herald Tribune*, February 4, 1932.
- Gelms, Joseph. "'Space Odyssey' Fails Most Gloriously." *Newsday*, April 4, 1968.
- Goertzel, Ben. "The Extropian Creed." 2000. Accessed February 28, 2026.
<https://www.goertzel.org/benzine/extropians.htm>.
- Great Britain. Inter-Departmental Committee on Physical Deterioration. *Report of the Inter-Departmental Committee on Physical Deterioration*. H.M.S.O., 1904.
http://archive.org/details/b32180366_0001.
- Haldane, J. B. S. *Daedalus or Science and the Future*. Kegan Paul, Trench, Trubner, 1924.
- Hanson, Robin. "The Great Filter — Are We Almost Past It?" 1998.
<https://mason.gmu.edu/~rhanson/greatfilter.html>.
- Huxley, Aldous. *Between the Wars: Essays and Letters*. Edited by David Bradshaw. I. R. Dee, 1994.
- Huxley, Aldous. *Brave New World*. Chatto & Windus, 1931.
- Huxley, Aldous. "The Outlook for American Culture." *Harper's Magazine*, August 1927.

- Huxley, Aldous. "What Is Happening to Our Population?" In *Debating Malthus*. University of Washington Press, 1934.
<https://www.degruyterbrill.com/document/doi/10.1515/9780295749914-032/html>.
- Huxley, Julian. *Essays of a Biologist*. Chatto & Windus, 1923.
- Huxley, Julian. *Memories*. Harper & Row, 1970.
- Huxley, Julian. *Memories II*. Harper & Row, 1973.
- Huxley, Julian. "UNESCO: Its Purpose and Philosophy." UNESCO, 1946.
<https://unesco.org.uk/resources/unesco-its-purpose-and-philosophy-julian-huxley-1946>.
- Huxley, Juliette. *Leaves of the Tulip Tree*. John Murray, 1986.
- K., K. "Brave New World." *Country Life* 71 (February 13, 1932): 182.
- Land, Nick. "The Dark Enlightenment." *Urban Future*, 2012.
<https://www.thedarkenlightenment.com/the-dark-enlightenment-by-nick-land/>.
- "Man of the Future." *Cincinnati Enquirer*, October 8, 1929. Vol. 86, Issue 281.
- McClay, Bob. "2001: A Space Odyssey." *Rolling Stone*, July 6, 1968.
- "Membership." Humanity+. Accessed March 1, 2026. <https://www.humanityplus.org/membership>.
- "MIT Extropians: Embrace the Future." Extropian pamphlet. Accessed via Scribd, March 16, 2026.
<https://www.scribd.com/document/250804968/Extropians>.
- More, Max. "Extropy #6." *Extropy*, 1990. <https://diyhpl.us/~bryan/irc/extropians/extropy-magazine/ext6.pdf>.
- More, Max. "Philosophy of Transhumanism." Humanity+. Accessed March 1, 2026.
<https://www.humanityplus.org/philosophy-of-transhumanism>.
- More, Max. "The Overhuman in the Transhuman." *Journal of Evolution and Technology* 21, no. 1 (2010). <https://jetpress.org/v21/more.pdf>.
- More, Max. "The Philosophy of Transhumanism." In *The Transhumanist Reader: Classical and Contemporary Reflections on the Science, Technology, and Philosophy of the Human Future*, edited by Max More and Natasha Vita-More, 3–17. Wiley-Blackwell, 2013.
- More, Max. "Transhumanism: Critical Rationalism, Not Omniscient Reason or Scientism." *Extropic Thoughts* (Substack), February 4, 2024. <https://maxmore.substack.com/p/transhumanism-critical-rationalism>.

Moravec, Hans P. *Mind Children: The Future of Robot and Human Intelligence*. Harvard University Press, 1988.

"Mr. Huxley's Horrible World." *The Bystander* 113 (February 17, 1932): 332.

Neuralink. "Two Years of Telepathy." *Neuralink Updates*, January 29, 2026. <https://neuralink.com/updates/two-years-of-telepathy/>.

"Peter Thiel Brings His Lectures on the Antichrist to the Vatican's Doorstep." Associated Press, March 16, 2026. <https://www.pbs.org/newshour/nation/peter-thiel-brings-his-lectures-on-the-antichrist-to-the-vaticans-doorstep>.

"Prehistory of Transhumanism." H+Pedia. Accessed November 14, 2025. https://hpluspedia.org/wiki/Prehistory_of_Transhumanism.

Russell, Bertrand. "H.G. Wells." In *Portraits from Memory and Other Essays*. Simon and Schuster, 1956.

Russell, Bertrand. *Icarus, or The Future of Science*. E. P. Dutton, 1924.

Stapledon, Olaf. *Story of the Near and Far Future: Last and First Men*. Penguin Books, 1937.

T. L. "Books of the Times: Strangled." *New York Times*, November 8, 1966.

Thiel, Peter. "The Education of a Libertarian." *Cato Unbound*, April 13, 2009. <https://www.cato-unbound.org/2009/04/13/peter-thiel/education-libertarian/>.

Thiel, Peter. Quoted in Emma Goldberg. "Peter Thiel is Delivering 4 Private Sold-Out Lectures in San Francisco — About the Antichrist." *Fortune*, September 2, 2025. <https://fortune.com/2025/09/02/peter-thiel-antichrist-lectures-commonwealth-club-san-francisco-acts-17-collective-rene-girard/>.

"The Ingenious Philosophic Romancer's Latest Imaginative Work." *New York Times*, 1905. BR342.

"Transhumanist FAQ." Humanity+. Accessed March 7, 2026. <https://www.humanityplus.org/transhumanist-faq>.

"Transhumanism Timeline: Significant Contributions Shaping the Evolution of Transhumanist Philosophy." Transhumanist Party. Accessed December 9, 2026. <https://transhumanist-party.org/transhumanism-timeline/>.

Walsh, Moira. "2001: A Space Odyssey." *America*, April 27, 1968.

Watters, Jim. "'2001: A Space Odyssey.'" *Boxoffice*, April 8, 1968.

Wells, H. G. *A Modern Utopia*. Chapman & Hall, 1905.

Wells, H. G. *Anticipations*. Chapman & Hall, 1901.

Wells, H. G. *Experiment in Autobiography*. Macmillan, 1934.

Wells, H. G. *The Secret Places of the Heart*. Macmillan, 1922.

Yarvin, Curtis. Quoted in Corey Pein. "Mouthbreathing Machiavellis Dream of a Silicon Reich." *The Baffler*, May 19, 2014. <https://thebaffler.com/salvos/mouthbreathing-machiavellis>.

Yudkowsky, Eliezer. *Rationality: A-Z*. LessWrong, n.d. Accessed February 17, 2026. <https://www.lesswrong.com/rationality>.

"2001: A Space Odyssey." *Monthly Film Bulletin*, 1968.

"Book Notes." *Hollywood Reporter* 188, no. 28 (1965): 6.

Secondary Sources

Abrams, Nathan. "2001: A Space Odyssey Still Leaves an Indelible Mark on Our Culture 55 Years On." *The Conversation*, July 10, 2023. <https://doi.org/10.64628/AB.7vkmygddh>.

Adelman, Paul. *The Rise of the Labour Party, 1880–1945*. 3rd ed. Routledge, 1996.

Bajarin, Tim. "Silicon Valley's Shifting Politics — and The One Thing Uniting Them." *Forbes*, May 20, 2025. <https://www.forbes.com/sites/timbajarin/2025/05/20/silicon-valleys-shifting-politics-and-the-one-thing-uniting-them/>.

Bashford, Alison. "Julian Huxley's Transhumanism." In *Crafting Humans: From Genesis to Eugenics and Beyond*, edited by Marius Turda, 153–68. V&R Unipress, 2013.

Becker, Adam. *More Everything Forever: AI Overlords, Space Empires, and Silicon Valley's Crusade to Control the Fate of Humanity*. Basic Books, 2025.

Begiato, Joanne. "Punishing the Unregulated Manly Body and Emotions in Early Victorian England." In *The Victorian Male Body*, edited by Joanne Ella Parsons and Ruth Heholt. Edinburgh University Press, 2018.

Bennett, Joshua. "A History of 'Rationalism' in Victorian Britain." *Modern Intellectual History* 15, no. 1 (2018): 63–91. <https://doi.org/10.1017/S1479244315000438>.

Bohan, Elise. *History of Transhumanism*. PhD diss., Macquarie University, 2022. <https://doi.org/10.25949/19444418.v1>.

Bostrom, Nick. "A History of Transhumanist Thought." 2005. <https://nickbostrom.com/papers/a-history-of-transhumanist-thought/>.

- Bostrom, Nick. "Transhumanist Values." *Journal of Philosophical Research* 30 (2005): 3.
- Bostrom, Nick. "Where Are They? Why I Hope the Search for Extraterrestrial Life Finds Nothing." *Technology Review*, May/June 2008, 72–77. <https://nickbostrom.com/papers/where-are-they/>.
- Bradshaw, David. *The Hidden Huxley: Contempt and Compassion for the Masses*. Faber and Faber, 1994.
- Brown, Andrew. *J.D. Bernal: The Sage of Science*. Oxford University Press, 2005.
- Buchanan, Tom. *The Impact of the Spanish Civil War on Britain: War, Loss and Memory*. Sussex Studies in Spanish History. Sussex Academic Press, 2007.
- Campbell, Colin. "Astrology." In *Encyclopedia of Religion and Society*, edited by William H. Swatos Jr. AltaMira Press, 1998. <https://hirr.hartfordinternational.edu/articles/4141/>.
- Cantor, Paul A. "The Empire of the Future: Imperialism and Modernism in H.G. Wells." paulcantor.io, May 24, 2020. <https://paulcantor.io/paul-cantor-works/the-empire-of-the-future-imperialism-and-modernism-in-hg-wells>.
- Carter, E. H. *A History of Britain, Volume VII: Liberal England, World War and Slump, 1901–1939*. Stacey International, 2011.
- Chafkin, Max. *The Contrarian: Peter Thiel and Silicon Valley's Pursuit of Power*. Penguin Press, 2021.
- Chamberlain, Fred. "Cryonics: FM-2030 Now Hurtling into the Future." *Cryonics* (Alcor Life Extension Foundation), 2000. <https://web.archive.org/web/20101119134918/http://alcor.org/cryonics/cryonics2000-4.pdf>.
- Chartier, Roger. *Cultural History: Between Practices and Representations*. Translated by Lydia Cochrane. Cornell University Press, 1988.
- Claiborne, Robert. "Stoned on Technology." *Washington Post, Times Herald*, March 7, 1971, 4.
- Clark, Ronald. *J.B.S.: The Life and Work of J.B.S. Haldane*. Oxford University Press, 1984.
- Cole, Margaret. "H.G. Wells and the Fabian Society." In *Edwardian Radicalism, 1900–1914: Some Aspects of British Radicalism*, edited by A. J. A. Morris. Routledge & Kegan Paul, 1974.
- Cohn, Victor. "Idea for New Space Movie Came From a Fertile Mind." *Washington Post*, March 31, 1968.
- Cooke, Barbara. "Professor David Bradshaw, 1955–2016." *Waugh and Words* (blog), September 2016. <https://staffblogs.le.ac.uk/waughandwords/2016/09/15/professor-david-bradshaw-1955-2016/>.
- Crossley, Robert. *Olaf Stapledon: Speaking for the Future*. Syracuse University Press, 1994.

- Dixon, Thomas. *Weeping Britannia: Portrait of a Nation in Tears*. Oxford University Press, 2015.
- Dunér, Ingrid. "Evolutionary Apocalypse." In *Julian Huxley, Evolutionism and the History of Transhumanism*, 165–97. Palgrave Macmillan, 2025.
- Dunér, Ingrid. "Huxley's Latest Word." In *Julian Huxley, Evolutionism and the History of Transhumanism*, 225–55. Palgrave Macmillan, 2025.
- Dunér, Ingrid. *Julian Huxley, Evolutionism and the History of Transhumanism*. Palgrave Studies in the Future of Humanity and Its Successors. Palgrave Macmillan, 2025. <https://doi.org/10.1007/978-3-031-81720-5>.
- Dvorsky, George, and James Hughes. *Postgenderism: Beyond the Gender Binary*. Institute for Ethics and Emerging Technologies, 2008. <https://doi.org/10.13140/RG.2.2.18028.08324>.
- Dyson, Freeman. *Disturbing the Universe*. Harper & Row, 1979.
- Erdmann, Gero, and Marianne Kneuer. *Regression of Democracy?* Zeitschrift für Vergleichende Politikwissenschaft. VS Verlag für Sozialwissenschaften, 2011. <https://doi.org/10.1007/978-3-531-93302-3>.
- Evans, Jules. "How Did Transhumanism Become the Religion of the Super-Rich?" *Philosophy for Life*, October 28, 2022. <https://www.philosophyforlife.org/blog/how-did-transhumanism-become-the-religion-of-the-super-rich>.
- Fanon, Frantz, and Constance Farrington. *The Wretched of the Earth*. Grove Press, 1961.
- Farman, Abou. "Transhumanism." In *The Open Encyclopedia of Anthropology*, edited by Felix Stein. 2022. <http://doi.org/10.29164/22transhumanism>.
- Firchow, Peter. *The End of Utopia: A Study of Aldous Huxley's Brave New World*. Bucknell University Press, 1984.
- Fisher, Allen, and Lyndon B. Johnson Presidential Library and Museum. "Women's Rights and the Civil Rights Act of 1964." National Archives, July 3, 2019. <https://www.archives.gov/women/1964-civil-rights-act>.
- Friedman, Seth. "Beyond the Infinite Interpretations: The Reception of 2001: A Space Odyssey and the Reframing of Stanley Kubrick's Authorial Reputation." *Film History* 32, no. 1 (2020): 72–105.
- Gahan, Peter. "1905–9: Noises Off — H. G. Wells Among the Fabians. Votes for Women!" In *Bernard Shaw and Beatrice Webb on Poverty and Equality in the Modern World, 1905–1914*. Palgrave Macmillan, 2017. https://doi.org/10.1007/978-3-319-48442-6_3.

- Geburu, Timnit, and Émile P. Torres. "The TESCREAL Bundle: Eugenics and the Promise of Utopia through Artificial General Intelligence." *First Monday*, April 14, 2024. <https://doi.org/10.5210/fm.v29i4.13636>.
- Geraci, Robert M. "There and Back Again: Transhumanist Evangelism in Science Fiction and Popular Science." *Implicit Religion* 14, no. 2 (2011): 141–47. <https://doi.org/10.1558/imre.v14i2.141>.
- Gilbert, Bentley B. "Health and Politics: The British Physical Deterioration Report of 1904." *Bulletin of the History of Medicine* 39, no. 2 (1965): 143–53.
- Goldberg, Emma. "Peter Thiel is Delivering 4 Private Sold-Out Lectures in San Francisco — About the Antichrist." *Fortune*, September 2, 2025. <https://fortune.com/2025/09/02/peter-thiel-antichrist-lectures-commonwealth-club-san-francisco-acts-17-collective-rene-girard/>.
- Goodman, Amy. "Alliance Between Far Right and Silicon Valley Gives Rise to 'End Times Fascism.'" *Truthout*, May 5, 2025. <https://truthout.org/video/alliance-between-far-right-and-silicon-valley-gives-rise-to-end-times-fascism/>.
- "The Greatest Books: The Modern Library | 100 Best Novels from Modern Library." Accessed October 14, 2025. <https://thegreatestbooks.org/lists/2>.
- "HG Wells — Father of Science Fiction with Hopes and Fears for How Science Will Shape Our Future." ABC Science Show, January 19, 2024. <https://www.abc.net.au/listen/programs/scienceshow/hg-wells-author-who-imaged-a-future-based-on-scientific-achievements/13731742>.
- Hagey, Keach. "How Peter Thiel's Relationship With Eliezer Yudkowsky Launched the AI Revolution." *Wired*, May 20, 2025. <https://www.wired.com/story/book-excerpt-the-optimist-open-ai-sam-altman/>.
- "History." London School of Economics. Accessed March 19, 2026. <https://www.lse.ac.uk/about-lse/our-history>.
- Hobsbawm, E. J. *The Age of Extremes: A History of the World, 1914–1991*. Vintage Books, 1996.
- Houghton, Walter E. *The Victorian Frame of Mind: 1830–1870*. Yale University Press, 1963.
- Hughes, James J. "The Politics of Transhumanism and the Techno-Millennial Imagination, 1626–2030." *Zygon* 47, no. 4 (2012): 757–76. <https://doi.org/10.1111/j.1467-9744.2012.01289.x>.
- Institute of Medicine (US), Steering Committee for the Symposium on the Medical Implications of Nuclear War. "Adult Beliefs, Feelings, and Actions Regarding Nuclear War: Evidence from Surveys and Experiments." In *The Medical Implications of Nuclear War*. National Academies Press, 1986. <https://www.ncbi.nlm.nih.gov/books/NBK219181/>.
- Jackson, Roy. *Muslim and Supermuslim: The Quest for the Perfect Being and Beyond*. Springer International Publishing, 2020. <https://doi.org/10.1007/978-3-030-37093-0>.

- Jackson, Roy. "Secular Transhumanism as Scientism?" In *Muslim and Supermuslim: The Quest for the Perfect Being and Beyond*, 231–70. Springer International Publishing, 2020. https://doi.org/10.1007/978-3-030-37093-0_3.
- Kaiser, David, and W. Patrick McCray. *Groovy Science: Knowledge, Innovation, and American Counterculture*. University of Chicago Press, 2019. <https://doi.org/10.7208/9780226373072-001>.
- Kriegel, Leonard. "Private & Public Sensibility." *Nation* 203, no. 19 (1966): 616–20.
- "Laureates." UNESCO. <https://www.unesco.org/en/prizes/popularization-science/laureates>.
- Leverage, Megan. "Worlds Apart? Julian Huxley, Transhumanism, and Eugenics." *Contemporanea* 21, no. 3 (2018): 431–58.
- Levin, Dan, and Craig Nigrelli. "Silicon Valley Leans Historically Left, so Why the Sudden Shift to the Right?" *Straight Arrow News*, August 15, 2025. <https://san.com/cc/silicon-valley-leans-historically-left-so-why-the-sudden-shift-to-the-right/>.
- Levin, Susan B. *Posthuman Bliss? The Failed Promise of Transhumanism*. Oxford University Press, 2021.
- Levin, Susan B. "Silicon Valley's Favorite Weird Philosophy Is Fundamentally Wrong." *Slate*, March 22, 2022. <https://slate.com/technology/2022/03/silicon-valley-transhumanism-eugenics-information.html>.
- Levine, Philippa. *Victorian Feminism, 1850–1900*. Hutchinson Education, 1987.
- Madden, Deanna. "Clockwork Orange." In *Misogyny in Literature: An Essay Collection*, edited by Katherine Anne Ackley. Garland Library, 1992.
- Milburn, Josephine Fishel. "The Fabian Society and the British Labour Party." *Western Political Quarterly* 11, no. 2 (1958): 319–39. <https://doi.org/10.2307/444409>.
- Molony, Barbara, and Jennifer Nelson, eds. *Women's Activism and "Second Wave" Feminism: Transnational Histories*. Bloomsbury Academic, 2017.
- "Monopoly Was Invented to Demonstrate the Evils of Capitalism." BBC, July 28, 2017. <https://www.bbc.com/worklife/article/20170728-monopoly-was-invented-to-demonstrate-the-evils-of-capitalism>.
- "Musk and Altman Lead a New Era of Transhumanism with Brain Implants and AI." BGNES, August 22, 2025. <https://www.bgnes.com/science/musk-and-altman-lead-a-new-era-of-transhumanism-with-brain-implants-and-ai>.
- Nkrumah, Kwame. *Neo-Colonialism: The Last Stage of Imperialism*. Panaf, 2004.

- O'Connell, Mark. *To Be a Machine*. Doubleday, 2017.
- O'Gieblyn, Meghan. "Ghost in the Cloud." *n+1*, April 2017. <https://www.nplusonemag.com/issue-28/essays/ghost-in-the-cloud/>.
- Ozmon, Howard. Review of *Optimism One: The Emerging Radicalism*, by F. M. Esfandiary. *Phi Delta Kappan* 52, no. 9 (1971): 562. <http://www.jstor.org/stable/20373016>.
- Parrinder, Patrick. *The Earth Is My Footstool: Wells, Stapledon and the Idea of the Post-Human*. January 2012.
- Pein, Corey. "Mouthbreathing Machiavellis Dream of a Silicon Reich." *The Baffler*, May 19, 2014. <https://thebaffler.com/salvos/mouthbreathing-machiavellis>.
- Pellissier, Hank. "Transhumanists: Who Are They? What Do They Want, Believe, and Predict? (Terasem Survey, Part 5)." Institute for Ethics and Emerging Technologies, September 9, 2012. <https://ieet.org/index.php/IEET2/more/pellissier20120909>.
- Perlstein, Rick. *The Invisible Bridge: The Fall of Nixon and the Rise of Reagan*. Simon & Schuster, 2014.
- Phillip, John Kennedy. "A Philosophical History of Transhumanism." *Philosophy Now*, February 2024. https://philosophynow.org/issues/160/A_Philosophical_History_of_Transhumanism.
- Robins-Early, Nick. "Oxford Shuts Down Institute Run by Elon Musk-Backed Philosopher." *The Guardian*, April 19, 2024. <https://www.theguardian.com/technology/2024/apr/19/oxford-future-of-humanity-institute-closes>.
- Russell, Calum. "Stanley Kubrick's '2001' Named Greatest Movie of All Time." *Far Out Magazine*, December 2, 2022. <https://faroutmagazine.co.uk/2001-a-space-odyssey-named-greatest-movie-of-all-time/>.
- Shane, Charlotte. "Life Extension Technology Gives Us a Bleak Future: More White Men." *Jezebel*, 2016. <https://www.jezebel.com/life-extension-technology-gives-us-a-bleak-future-more-1793857274>.
- Smith, David Clayton. *H.G. Wells: Desperately Mortal: A Biography*. Yale University Press, 1986.
- Sorensen, Clark. "The Evolutionary Humanism of Julian Huxley and the Creation of UNESCO." *Journal of the History of Ideas* 60, no. 3 (1999): 531–53.
- Taplin, Jonathan. "How Musk, Thiel, Zuckerberg, and Andreessen — Four Billionaire Techno-Oligarchs — Are Creating an Alternate, Autocratic Reality." *Vanity Fair*, August 22, 2023. <https://www.vanityfair.com/news/2023/08/musk-thiel-zuckerberg-andreessen-alternate-autocratic-reality>.

- Taylor, J. H. C. "Bio-Medical Degeneration in Britain, 1850–1914: Theory and Evidence." DPhil thesis, University of Oxford, 2021.
- Thatcher, David S. *Nietzsche in England 1890–1914: The Growth of a Reputation*. University of Toronto Press, 1970.
- Thomas, Alexander. "A Brief History of Transhumanism and Its Critics." In *The Politics and Ethics of Transhumanism*. Bristol University Press, 2024. <https://doi.org/10.2307/jj.14284473.4>.
- Tirosh-Samuelson, Hava. "Transhumanism as a Secularist Faith." *Zygon* 47, no. 4 (2012). <https://doi.org/10.1111/j.1467-9744.2012.01288.x>.
- Tiven, Benjamin. "The Future Takes Forever: Becoming FM-2030." *Bidoun*, n.d. <https://bidoun.org/articles/the-future-takes-forever>.
- Vita-More, Natasha. "History of Transhumanism." In *The Transhumanism Handbook*, edited by Newton Lee. Springer International Publishing, 2019. https://doi.org/10.1007/978-3-030-16920-6_2.
- Weindling, Paul. "Julian Huxley and the Continuity of Eugenics in Twentieth-Century Britain." *Journal of Modern European History* 10, no. 4 (2012): 480–99. https://doi.org/10.17104/1611-8944_2012_4_480.
- Werskey, Gary. *The Visible College: The Collective Biography of British Scientific Socialists of the 1930s*. Allen Lane, 1978.
- Woiak, Joanne. "Designing a Brave New World: Eugenics, Politics, and Fiction." *The Public Historian* 29, no. 3 (2007): 105–129.
- Wolyniak, Joseph, and Peter Harrison. "History of 'Transhumanism.'" *Notes and Queries* 62, no. 1 (2015): 138–39. <https://doi.org/10.1093/notesj/gjv080>.
- Wuyangga. "The Political and Economic Impact of the Long Depression (1873–1896) on Late Victorian Britain." *Journal of Applied Economics and Policy Studies* 18, no. 7 (2025): 53–61. <https://doi.org/10.54254/2977-5701/2025.25766>.